

Chapter 17

Continuity and Differentiability

Solutions

SECTION - A

Objective Type Questions (One option is correct)

1. If the function $f(x) = (1-x)\tan\frac{\pi x}{2}$ is continuous at $x = 1$, then $f(1) =$
- (1) $\frac{2}{\pi}$ (2) $\frac{\pi}{2}$ (3) 0 (4) 2π

Sol. Answer (1)

$$\begin{aligned} f(1) &= \lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} (1-x)\tan\left(\frac{\pi x}{2}\right) \\ &= \lim_{h \rightarrow 0} (1-(1+h))\tan\left(\frac{\pi}{2}(1+h)\right) \\ &= \lim_{h \rightarrow 0} (-h)\tan\left(\frac{\pi}{2} + \frac{\pi h}{2}\right) \\ &= \lim_{h \rightarrow 0} (-h)\left(-\cot\frac{\pi h}{2}\right) \\ &= \lim_{h \rightarrow 0} \frac{\frac{\pi}{2} \cdot h}{\tan\left(\frac{\pi h}{2}\right)} \cdot \frac{2}{\pi} = \frac{2}{\pi} \end{aligned}$$

2. Let $f(x) = \begin{cases} x \sin\left(\frac{1}{x}\right), & x \neq 0 \\ k, & x = 0 \end{cases}$

then $f(x)$ is continuous at $x = 0$, if

- (1) $k = 1$ (2) $k = 0$ (3) $k = 2$ (4) $k = -1$

Sol. Answer (2)

$$\lim_{x \rightarrow 0} x \sin \frac{1}{x} = 0$$

$\therefore k = 0$ will make f continuous at 0.

$$(\because \lim_{x \rightarrow 0} (x) = 0 \text{ and } \sin\left(\frac{1}{x}\right) \leq 1 \text{ for } x \neq 0)$$

3. Let $f(x) = \begin{cases} \frac{3|x| + 4 \tan x}{x}, & x \neq 0 \\ k, & x = 0 \end{cases}$

then $f(x)$ is continuous at $x = 0$ for

- (1) $k = 7$ (2) $k = 1$ (3) No values of k (4) $k = 2$

Sol. Answer (3)

$$\begin{aligned} \lim_{x \rightarrow 0^-} f(x) &= \lim_{x \rightarrow 0^-} \frac{3|x| + 4 \tan x}{x} \\ &= \lim_{x \rightarrow 0^-} \frac{3(-x) + 4 \tan x}{x} \\ &= \lim_{x \rightarrow 0^-} \frac{-3x}{x} + \lim_{x \rightarrow 0^-} \frac{4 \tan x}{x} \\ &= -3 + 4 = 1 \end{aligned}$$

$$\begin{aligned} \text{and } \lim_{x \rightarrow 0^+} f(x) &= \lim_{x \rightarrow 0^+} \frac{3|x| + 4 \tan x}{x} \\ &= \lim_{x \rightarrow 0^+} \left(\frac{3x}{x} + \frac{4 \tan x}{x} \right) \\ &= 3 + 4 = 7 \end{aligned}$$

$$\lim_{x \rightarrow 0^-} f(x) \neq \lim_{x \rightarrow 0^+} f(x)$$

$\therefore f(x)$ is not continuous at $x = 0$, whatever k may be.

4. Let $f(x) = \begin{cases} x + a, & x < 1 \\ ax^2 + 1, & x \geq 1 \end{cases}$ then $f(x)$ is continuous at $x = 1$ for

- (1) $a = 0$ (2) $a = 1$ (3) All $a \in \mathbb{R}$ (4) No value of a

Sol. Answer (3)

$$f(1) = a + 1$$

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (x + a) = (1 + a)$$

$$\text{and } \lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} (ax^2 + 1) = a + 1$$

$\therefore f(x)$ is continuous at $x = 1$, whatever 'a' may be.

5. If the function $f(x) = \frac{x(e^{\sin x} - 1)}{1 - \cos x}$ is continuous at $x = 0$, then $f(0) =$

- (1) 1 (2) 0 (3) 2 (4) $\frac{1}{2}$

Sol. Answer (3)

$$\begin{aligned}
 f(0) = \lim_{x \rightarrow 0} f(x) &= \lim_{x \rightarrow 0} \frac{x(e^{\sin x} - 1)}{1 - \cos x} \\
 &= \lim_{x \rightarrow 0} \frac{e^{\sin x} - 1}{\sin x} \cdot \frac{\sin x}{x} \cdot \frac{x^2}{1 - \cos x} = 2
 \end{aligned}$$

6. Let $f(x) = \frac{x(2^x - 1)}{1 - \cos x}$ for $x \neq 0$. What choice of $f(0)$, if any, will make $f(x)$ continuous at $x = 0$?

- (1) $\log 2$ (2) $\frac{1}{2} \log 2$ (3) $\frac{1}{2 \log 2}$ (4) $2 \log 2$

Sol. Answer (4)

$$\begin{aligned}
 \lim_{x \rightarrow 0} f(x) &= \lim_{x \rightarrow 0} \frac{x(2^x - 1)}{1 - \cos x} \\
 &= \lim_{x \rightarrow 0} \frac{2^x - 1}{x} \cdot \frac{x^2}{1 - \cos x} \\
 &= (\log 2) \cdot \left(\frac{1}{\frac{1}{2}} \right) = 2 \log 2
 \end{aligned}$$

7. If $f(x) = \begin{cases} [x] + [-x] & , x \neq 0 \\ \lambda & , x = 0 \end{cases}$, where $[.]$ denotes the greatest integer function, then $f(x)$ is continuous at $x = 0$,

for $\lambda =$

- (1) -1 (2) 0 (3) 1 (4) No value is possible

Sol. Answer (1)

$$\begin{aligned}
 \lim_{x \rightarrow 0^-} f(x) &= \lim_{x \rightarrow 0^-} [x] + [-x] \\
 &= \lim_{x \rightarrow 0^-} [x] + \lim_{x \rightarrow 0^-} [-x] \\
 &= -1 + 0 = -1
 \end{aligned}$$

$$\begin{aligned}
 \lim_{x \rightarrow 0^+} f(x) &= \lim_{x \rightarrow 0^+} [x] + [-x] \\
 &= \lim_{x \rightarrow 0^+} [x] + \lim_{x \rightarrow 0^+} [-x] \\
 &= 0 + (-1) = -1
 \end{aligned}$$

$\therefore \lambda = -1$ will make f continuous at $x = 0$.

8. Let $f(x) = \begin{cases} \frac{\sin[x]}{[x]}, & [x] \neq 0 \\ 0, & [x] = 0 \end{cases}$, then $f(x)$ is ([.] denotes the greatest integer function)

(1) Continuous at $x = 0$

(2) Continuous at $x = 1$

(3) Discontinuous at $x = 0$

(4) Continuous at $x = -1$

Sol. Answer (3)

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} 0 = 0 = f(0)$$

$$\begin{aligned} \lim_{x \rightarrow 0^-} f(x) &= \lim_{x \rightarrow 0^-} \frac{\sin[x]}{[x]} \\ &= \lim_{x \rightarrow 0^-} \frac{\sin(-1)}{-1} \\ &= \sin 1 \neq f(0) \end{aligned}$$

9. Let $f(x) = \sin \frac{1}{x}$, $x \neq 0$. Then $f(x)$ can be continuous at $x = 0$

(1) If $f(0) = 1$

(2) If $f(0) = 0$

(3) If $f(0) = -1$

(4) For no value of $f(0)$

Sol. Answer (4)

For continuity at $x = 0$,

$$\lim_{h \rightarrow 0} \sin \frac{1}{0+h} = \lim_{h \rightarrow 0} \sin \frac{1}{0-h} = f(0)$$

But $\lim_{h \rightarrow 0} \sin \frac{1}{h} = \lim_{h \rightarrow 0} (\text{a number between } 1 \text{ and } -1)$

$\neq$ a definite number

$\therefore f(0)$ cannot be a definite number.

10. If $f(x) = \begin{cases} px^2 - q, & x \in [0, 1) \\ x+1, & x \in (1, 2] \end{cases}$

and $f(1) = 2$, then the value of the pair (p, q) for which $f(x)$ cannot be continuous at $x = 1$ is

(1) $(2, 0)$

(2) $(1, -1)$

(3) $(4, 2)$

(4) $(1, 1)$

Sol. Answer (4)

$f(x)$ is continuous at $x = 1$, if

$$\lim_{h \rightarrow 0} f(1+h) = \lim_{h \rightarrow 0} f(1-h) = f(1) = 2$$

$$\lim_{h \rightarrow 0} f(1+h) = \lim_{h \rightarrow 0} \{1+h+1\} = 2$$

$$\lim_{h \rightarrow 0} f(1-h) = \lim_{h \rightarrow 0} \{p(1-h)^2 - q\} = p - q$$

$\therefore p - q \neq 2$

$\therefore (1, 1)$ cannot be the solution.

11. If $f(x)$ is derivable at $x = 1$, then $\lim_{x \rightarrow a} \frac{xf(a) - af(x)}{x - a}$

- (1) $f'(a)$ (2) $af'(a) - f(a)$ (3) $f(a) - af'(a)$ (4) $af'(a)$

Sol. Answer (3)

$$\begin{aligned} & \lim_{x \rightarrow a} \frac{xf(a) - af(x)}{x - a} \\ &= \lim_{h \rightarrow 0} \frac{(a+h)f(a) - af(a+h)}{h} \\ &= \lim_{h \rightarrow 0} \frac{hf(a)}{h} + \lim_{h \rightarrow 0} \frac{a[f(a) - f(a+h)]}{h} \\ &= f(a) - a \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = f(a) - af'(a) \end{aligned}$$

12. Let $f(x) = x|x|$, then $f'(0)$ is equal to

- (1) 1 (2) -1 (3) 0 (4) ± 1

Sol. Answer (3)

$$f(x) = x|x| = \begin{cases} x^2 & ; x \geq 0 \\ -x^2 & ; x < 0 \end{cases}$$

$$\begin{aligned} Lf'(0) &= \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{-h} \\ &= \lim_{h \rightarrow 0} \left(\frac{-h^2 - 0^2}{-h} \right) = 0 \end{aligned}$$

$$\begin{aligned} Rf'(0) &= \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{h^2 - 0^2}{h} = 0 \end{aligned}$$

$$\Rightarrow Lf'(0) = Rf'(0)$$

$$\Rightarrow f'(0) = 0$$

13. Let $f(x) = \begin{cases} x+a & , x \geq 1 \\ ax^2+1 & , x < 1 \end{cases}$ then $f(x)$ is derivable at $x = 1$, if

- (1) $a = 1$ (2) $a = 0$ (3) $a = 2$ (4) $a = \frac{1}{2}$

Sol. Answer (4)

$$f(1) = a + 1$$

$$Lf'(1) = \lim_{h \rightarrow 0} \frac{f(1-h) - f(1)}{-h} = \lim_{h \rightarrow 0} \frac{a(1-h)^2 + 1 - (1+a)}{-h} = 2a$$

$$\text{and } Rf'(1) = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} = \lim_{h \rightarrow 0} \frac{1+h+a - (1+a)}{h} = 1$$

$$\therefore f(x) \text{ is derivable at } x = 1, \text{ only if } 2a = 1 \text{ i.e. if } a = \frac{1}{2}.$$

14. If $f(x) = e^{-\frac{1}{x^2}}$, $x \neq 0$ and $f(0) = 0$ then $f'(0)$ is

(1) Not defined

(2) 1

(3) e

(4) 2

Sol. Answer (1)

$$Rf'(0) = \lim_{h \rightarrow 0} \frac{e^{\frac{-1}{(0+h)^2}} - 0}{h} = \lim_{h \rightarrow 0} \frac{1}{h} e^{-\frac{1}{h^2}}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{1}{h}}{e^{\frac{1}{h^2}}}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{1}{h^2}}{e^{\frac{1}{h^2}} \cdot \left(-\frac{2}{h^3}\right)}$$

$$= \lim_{h \rightarrow 0} \frac{h}{2e^{\frac{1}{h^2}}} = \frac{0}{\infty} = 0$$

$$Lf'(0) = \lim_{h \rightarrow 0} \frac{e^{\frac{1}{(0-h)^2}} - 0}{-h} = \lim_{h \rightarrow 0} \left(-\frac{1}{h}\right) e^{\frac{1}{h^2}}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{1}{h}}{e^{\frac{1}{h^2}}}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{1}{h^2}}{e^{\frac{1}{h^2}} \cdot \left(-\frac{2}{h^3}\right)}$$

$$= \lim_{h \rightarrow 0} \frac{h}{-2e^{\frac{1}{h^2}}} = \frac{0}{\infty} = 0$$

15. If $f(x) = \log |x|$ then for $x \neq 0$, $f'(x)$ is equal to

(1) $\frac{1}{|x|}$ (2) $\frac{1}{x}$ (3) $-\frac{1}{x}$ (4) $\pm \frac{1}{x}$

Sol. Answer (2)

$$f'(x) = \frac{d}{dx}(\log |x|) = \frac{1}{|x|} \frac{d}{dx} |x|$$

$$= \frac{1}{|x|} \cdot \frac{x}{|x|} = \frac{x}{|x|^2} = \frac{1}{x}, x \neq 0$$

16. $\frac{d}{dx} \left(\sin^{-1} \frac{2x}{1+x^2} \right)$ is equal to

- (1) $\frac{2}{1+x^2}$ (2) $-\frac{2}{1+x^2}$ (3) $\frac{3(1-x^2)}{|1-x^2|(1+x^2)}, x \neq 1$ (4) $\frac{2}{1-x^2}$

Sol. Answer (1)

Put $x = \tan \theta$

$$\therefore f(x) = \sin^{-1} \frac{2 \tan \theta}{1 + \tan^2 \theta} = 2\theta = 2 \tan^{-1} x$$

$$\therefore f'(x) = \frac{2}{1+x^2}$$

17. Differential coefficient of $\log_{10} x$ w.r.t. $\log_x 10$ is

- (1) $-\frac{(\log x)^2}{(\log 10)^2}$ (2) $\frac{(\log_{10} x)^2}{(\log 10)^2}$ (3) $\frac{(\log_x 10)^2}{(\log 10)^2}$ (4) $-\frac{(\log 10)^2}{(\log x)^2}$

Sol. Answer (1)

$$\text{If } y = \log_{10} x \quad z = \log_x 10$$

$$y = \frac{\log x}{\log 10} \quad z = \frac{\log 10}{\log x}$$

$$\begin{aligned} \frac{dy}{dz} &= \frac{\frac{d}{dx} \left(\frac{\log x}{\log 10} \right)}{\frac{d}{dx} \left(\frac{\log 10}{\log x} \right)} = \frac{\left(\frac{1}{\log 10} \right) \cdot \frac{1}{x}}{(\log 10) \cdot \frac{-1}{(\log x)^2} \cdot \frac{1}{x}} \\ &= -\frac{(\log x)^2}{(\log 10)^2} \end{aligned}$$

18. If $y = \log \left[e^x \left(\frac{x-2}{x+2} \right)^{\frac{3}{4}} \right]$ then $\frac{dy}{dx} =$

- (1) $\frac{x^2-4}{x^2-1}$ (2) $\frac{3}{x^2-4}$ (3) $\frac{x^2-1}{x^2-4}$ (4) $\frac{3x^2}{x^2-4}$

Sol. Answer (3)

$$\begin{aligned} y &= \log \left(e^x \left(\frac{x-2}{x+2} \right)^{\frac{3}{4}} \right) \\ &= \log e^x + \frac{3}{4} [\log(x-2) - \log(x+2)] \\ &= x + \frac{3}{4} [\log(x-2) - \log(x+2)] \\ \frac{dy}{dx} &= 1 + \frac{3}{4} \left[\frac{1}{x-2} - \frac{1}{x+2} \right] \\ &= \frac{x^2-1}{x^2-4} \end{aligned}$$

19. If $y = \frac{e^x - e^{-x}}{e^x + e^{-x}}$, then $\frac{dy}{dx} =$

- (1) $1 + y^2$ (2) $1 - y^2$ (3) $y^2 - 1$ (4) $y^2 + 2$

Sol. Answer (2)

$$\frac{dy}{dx} = \frac{(e^x + e^{-x})(e^x + e^{-x}) - (e^x - e^{-x})(e^x - e^{-x})}{(e^x + e^{-x})^2} = 1 - y^2$$

20. If $y = ae^{mx} + be^{-mx}$, then $\frac{d^2y}{dx^2} =$

- (1) m^2y (2) $-m^2y^2$ (3) my (4) $-my$

Sol. Answer (1)

$$y_1 = ame^{mx} + b(-m)e^{-mx}$$

$$y_2 = am^2e^{mx} + b(-m)^2e^{-mx} = m^2y$$

21. If $y^2 = ax^2 + b$, then $\frac{d^2y}{dx^2} =$

- (1) $\frac{ab}{x^3}$ (2) $\frac{x^3}{ab}$ (3) $\frac{ab}{y^2}$ (4) $\frac{ab}{y^3}$

Sol. Answer (4)

$$y^2 = ax^2 + b$$

$$2y \frac{dy}{dx} = 2ax \Rightarrow \frac{dy}{dx} = \frac{ax}{y}$$

$$\frac{d^2y}{dx^2} = \frac{ay - ax \frac{dy}{dx}}{y^2} = \frac{ay - ax \left(\frac{ax}{y} \right)}{y^2} = \frac{ab}{y^3}$$

22. If $y = \frac{\log x}{x}$, then $\frac{d^2y}{dx^2} =$

- (1) $\frac{3 - 2\log x}{x^3}$ (2) $\frac{2\log x - 3}{x^3}$ (3) $\frac{2\log x - 3}{x^4}$ (4) $\frac{2 - 3\log x}{x^4}$

Sol. Answer (2)

$$y_1 = \frac{d}{dx} \left(\frac{\log x}{x} \right) = \frac{\left(\frac{1}{x} \right) x - \log x}{x^2} = \frac{1 - \log x}{x^2}$$

$$y_2 = \frac{x^2 \left(-\frac{1}{x} \right) - (1 - \log x) 2x}{(x^2)^2} = \frac{2\log x - 3}{x^3}$$

23. The differential coefficient of $f(\log_e x)$ w.r.t. x , where $f(x) = \log_e x$, is

- (1) $\frac{x}{\log_e x}$ (2) $\frac{1}{x} \log_e x$ (3) $\frac{1}{x \log_e x}$ (4) $x \log x$

Sol. Answer (3)

$$f(\log_e x) = \log_e (\log_e x)$$

$$\therefore \frac{df(\log_e x)}{dx} = \frac{1}{\log_e x} \cdot \frac{1}{x}$$

24. If $y = (1+x)(1+x^2)(1+x^4)\dots(1+x^{2^n})$ then $\frac{dy}{dx}$ at $x = 0$ is

(1) 1

(2) -1

(3) 0

(4) 2

Sol. Answer (1)

$$y = \frac{(1-x)(1+x)(1+x^2)(1+x^4)\dots(1+x^{2^n})}{(1-x)} = \frac{1-x^{2^{n+1}}}{1-x}$$

$$\therefore \frac{dy}{dx} = \frac{-2^{n+1} \cdot x^{2^{n+1}-1} \cdot (1-x) + 1 - x^{2^{n+1}}}{(1-x)^2}$$

at $x = 0$

$$\therefore \frac{dy}{dx} = \frac{-2^{n+1} \cdot 0 \cdot 1 + 1 - 0}{1^2} = 1.$$

25. If $f(x) = \cos x \cdot \cos 2x \cdot \cos 4x \cdot \cos 8x \cdot \cos 16x$ then $f'\left(\frac{\pi}{4}\right)$ is

(1) $\sqrt{2}$ (2) $\frac{1}{\sqrt{2}}$

(3) 1

(4) -1

Sol. Answer (1)

$$f(x) = \frac{2 \sin x \cos x \cos 2x \cos 4x \cos 8x \cos 16x}{2 \sin x} = \frac{\sin 32x}{2^5 \sin x}$$

$$f'(x) = \frac{1}{32} \frac{32 \cos 32x \sin x - \cos x \sin 32x}{\sin^2 x}$$

$$f'\left(\frac{\pi}{4}\right) = \frac{32 \cdot \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}(0)}{32 \cdot \left(\frac{1}{\sqrt{2}}\right)^2}$$

26. If $y = \cos^{-1}(\cos x)$ then $\frac{dy}{dx}$ at $x = \frac{5\pi}{4}$ is equal to

(1) 1

(2) -1

(3) $\frac{1}{\sqrt{2}}$ (4) $\sqrt{2}$ **Sol.** Answer (2)

$$y = \cos^{-1} \cos(\pi + x - \pi)$$

$$= \cos^{-1} \cos(\pi - x - \pi)$$

When x is around $\frac{5\pi}{4}$, $\pi - x - \pi$ is in second quadrant.

$$\text{so } y = \pi - (x - \pi)$$

$$\therefore \frac{dy}{dx} = -1.$$

27. If $x = e^{y+e^y+e^{y+\dots\infty}}$ then $\frac{dy}{dx}$ is

(1) $\frac{x}{1+x}$

(2) $\frac{1}{x}$

(3) $\frac{1-x}{x}$

(4) $\frac{x}{1-x}$

Sol. Answer (3)

$$x = e^{y+e^y+e^{y+\dots\infty}}$$

$$\log x = y + x$$

$$\frac{1}{x} = \frac{dy}{dx} + 1$$

28. If $x^y \cdot y^x = 16$ then $\frac{dy}{dx}$ at (2, 2) is equal to

(1) 1

(2) -1

(3) 0

(4) 2

Sol. Answer (2)

$$y \log x + x \log y = \log 16$$

$$\frac{y}{x} + (\log x) \frac{dy}{dx} + (\log y) + \frac{x}{y} \frac{dy}{dx} = 0$$

$$1 + (1) \frac{dy}{dx} = 0$$

$$\therefore \left. \frac{dy}{dx} \right|_{(2,2)} = -1.$$

29. If $y = \sin x^\circ$ and $z = \cos x$, then $\frac{dy}{dz}$ is equal to

(1) $-\operatorname{cosec} x \cos x$

(2) $\frac{\pi}{180} \operatorname{cosec} \frac{\pi x}{180} \cos x$

(3) $-\frac{\pi}{180} \operatorname{cosec} x \cos \frac{\pi x}{180}$

(4) $\frac{\pi}{180} \sin x \frac{\cos \pi x}{180}$

Sol. Answer (3)

$$y = \sin \frac{x\pi}{180}; z = \cos x$$

$$\frac{dy}{dz} = \frac{\cos \frac{\pi x}{180} \cdot \frac{\pi}{180}}{-\sin x} = \frac{\pi}{180} \frac{\cos \pi x}{180} (-\operatorname{cosec} x).$$

30. Let the function $y = f(x)$ be given by

$$x = t^5 - 5t^3 - 20t + 7$$

$$\text{and } y = 4t^3 - 3t^2 - 18t + 3,$$

where $t \in (-2, 2)$ then $f'(x)$ at $t = 1$ is

(1) $\frac{5}{2}$

(2) $\frac{2}{5}$

(3) $\frac{7}{5}$

(4) $\frac{5}{7}$

Sol. Answer (2)

$$\left. \frac{dy}{dx} \right|_{t=1} = \left. \frac{12t^2 - 6t - 18}{5t^4 - 15t^2 - 20} \right|_{t=1} = \frac{12 - 6 - 18}{5 - 15 - 20} = \frac{2}{5}.$$

31. If $u = f(x^3)$; $v = g(x^2)$; $f'(x) = \cos x$ and $g'(x) = \sin x$ $\frac{du}{dv}$ is

- (1) $\frac{3}{2}x \cos x^3 \operatorname{cosec} x^2$ (2) $\frac{2}{3} \sin x^3 \sec x^2$ (3) $\tan x$ (4) $\frac{3}{2}x \sin x^2 \sec x^3$

Sol. Answer (1)

$$\frac{du}{dv} = \frac{\frac{du}{dx}}{\frac{dv}{dx}} = \frac{f'(x^3) \cdot 3x^2}{g'(x^2) \cdot 2x} = \frac{\cos x^3 \cdot 3x^2}{\sin x^2 \cdot 2x} = \frac{3}{2}x \cos x^3 \operatorname{cosec} x^2.$$

32. The differential coefficient of $\operatorname{cosec}^{-1} \frac{1}{2x^2 - 1}$ w.r.t. $\sqrt{1 - x^2}$ at $x = \frac{1}{2}$ is equal to

- (1) -4 (2) 4 (3) -1 (4) -2

Sol. Answer (1)

$$x = \cos \theta$$

$$y = \operatorname{cosec}^{-1} \frac{1}{2x^2 - 1} = \operatorname{cosec}^{-1} \sec 2\theta$$

$$= \operatorname{cosec}^{-1} \left\{ \operatorname{cosec} \left(\frac{\pi}{2} - 2\theta \right) \right\} = \frac{\pi}{2} - 2\theta$$

$$z = \sqrt{1 - \cos^2 \theta} = \sin \theta$$

$$\therefore \frac{dy}{dz} = \frac{\frac{dy}{d\theta}}{\frac{dz}{d\theta}} = \frac{-2}{\cos \theta} = \frac{-2}{x}$$

$$\therefore \left. \frac{-2}{x} \right|_{x=\frac{1}{2}} = -4.$$

33. If $t(1 + x^2) = x$ and $x^2 + t^2 = y$ then $\frac{dy}{dx}$ at $x = 2$ is

- (1) $\frac{88}{125}$ (2) $\frac{488}{125}$ (3) 1 (4) $\frac{244}{125}$

Sol. Answer (2)

$$t = \frac{x}{1 + x^2}$$

$$\text{and } 2x + 2t \frac{dt}{dx} = \frac{dy}{dx}$$

$$\therefore \frac{dt}{dx} = \frac{(1 + x^2) - x(2x)}{(1 + x^2)^2} = \frac{1 - x^2}{(1 + x^2)^2}$$

$$\therefore \frac{dy}{dx} = 2x + \frac{2x}{1+x^2} \cdot \frac{1-x^2}{(1+x^2)^2}$$

Now, put $x = 2$

$$\Rightarrow \frac{dy}{dx} = \frac{488}{125}$$

34. $x = t \cos t$, $y = t + \sin t$ then $\frac{d^2x}{dy^2}$ at $\left(t = \frac{\pi}{2}\right)$ is equal to

(1) $\frac{\pi+4}{2}$

(2) $-\frac{\pi+4}{2}$

(3) -2

(4) 2

Sol. Answer (2)

$$\frac{dx}{dy} = \frac{\frac{dx}{dt}}{\frac{dy}{dt}} = \frac{\cos t - t \sin t}{1 + \cos t}$$

$$\therefore \frac{d^2x}{dy^2} = \frac{\frac{d}{dt}\left(\frac{dx}{dy}\right)}{\frac{dy}{dt}} = \frac{(-2 \sin t - t \cos t)(1 + \cos t) - (\cos t - t \sin t)(-\sin t)}{(1 + \cos t)^2}$$

$$\text{Now, Put } t = \frac{\pi}{2} \Rightarrow \frac{d^2x}{dy^2} = -\frac{\pi+4}{2}$$

35. If $y = \sin 2x$, then $\frac{d^6y}{dx^6}$ at $x = \frac{\pi}{2}$ is equal to

(1) -64

(2) 0

(3) 64

(4) 128

Sol. Answer (2)

$$\frac{dy}{dx} = 2 \cos 2x = 2 \sin\left(2x + \frac{\pi}{2}\right)$$

$$\frac{d^2y}{dx^2} = 2^2 \cos\left(2x + \frac{\pi}{2}\right) = 2^2 \sin\left(2x + 2 \cdot \frac{\pi}{2}\right)$$

$$\frac{d^6y}{dx^6} = 2^6 \sin\left(2x + 6 \cdot \frac{\pi}{2}\right) = 2^6 \sin(2x + 3\pi)$$

$$\frac{d^6y}{dx^6} = 0 \text{ at } x = \frac{\pi}{2}$$

36. Let $f(x) = \begin{vmatrix} x^3 & \sin x & \cos x \\ 6 & -1 & 0 \\ p & p^2 & p^3 \end{vmatrix}$ where p is a constant. Then $\left.\frac{d^3}{dx^3}f(x)\right|_{x=0}$ is

(1) p

(2) $p + p^2$

(3) $p + p^3$

(4) Independent of p

Sol. Answer (4)

$$f(x) = -p^3x^3 - 6p^3 \sin x + (6p^2 + p)\cos x$$

$$f'(x) = -3p^3 x^2 - 6p^3 \cos x - (6p^2 + p) \sin x$$

$$f''(x) = -6p^3 x + 6p^3 \sin x - (6p^2 + p) \cos x$$

$$f'''(x) = -6p^3 + 6p^3 \cos x + (6p^2 + p) \sin x$$

$$f'''(0) = -6p^3 + 6p^3 = 0 \text{ (independent of } p\text{)}.$$

37. If $y = at^2$, $x = 2at$, where a is a constant, then $\frac{d^2y}{dx^2}$ at $x = \frac{1}{2}$ is equal to

- (1) $8a$ (2) 1 (3) $2a$ (4) $\frac{1}{2a}$

Sol. Answer (4)

The given function can be written as

$$x^2 = 4ay$$

$$\Rightarrow \frac{dy}{dx} = \frac{x}{2a} \Rightarrow \frac{d^2y}{dx^2} = \frac{1}{2a}$$

38. If $y = x^{\frac{1}{x}}$, the value of $\frac{dy}{dx}$ at $x = e$ is equal to

- (1) 1 (2) 0 (3) -1 (4) $e^{\left(\frac{1}{e}\right)-2}$

Sol. Answer (2)

$$y = x^{\frac{1}{x}}$$

$$\log y = \frac{1}{x} \log x$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{1}{x^2} + \log x \cdot \left(\frac{-1}{x^2}\right)$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{1}{e^2} + (1) \left(\frac{-1}{e^2}\right) = 0$$

$$\therefore \frac{dy}{dx} = 0.$$

39. If $y = \tan^{-1} \sqrt{\frac{x+1}{x-1}}$ for $|x| > 1$ then $\frac{dy}{dx} =$

- (1) $\frac{-1}{2|x|\sqrt{x^2-1}}$ (2) $\frac{-1}{2x\sqrt{x^2-1}}$ (3) $\frac{1}{2x\sqrt{x^2-1}}$ (4) $\frac{1}{2|x|\sqrt{x^2-1}}$

Sol. Answer (1)

Let $x = \sec \theta$

$$y = \tan^{-1} \sqrt{\frac{\sec \theta + 1}{\sec \theta - 1}}$$

$$= \tan^{-1} \sqrt{\frac{1 + \cos \theta}{1 - \cos \theta}}$$

$$= \tan^{-1} \cot\left(\frac{\theta}{2}\right)$$

$$= \frac{\pi}{2} - \frac{1}{2} \sec^{-1} x$$

$$\therefore \frac{dy}{dx} = -\frac{1}{2|x|\sqrt{x^2-1}}$$

40. If $y = \log_2 \log_e x$ then $\frac{dy}{dx}$ is equal to

- (1) $\frac{1}{x} \log_2 e \cdot \log_x e$ (2) $\frac{1}{x} \log_2 x$ (3) $\frac{1}{x} \log_e x$ (4) $\frac{1}{x} \log_x e$

Sol. Answer (1)

$$y = \frac{\log_e \log_e x}{\log_e 2} = \frac{\left(\frac{1}{x}\right) \cdot \frac{1}{\log_e x}}{\log_e 2} = \frac{1}{x} \log_2 e \log_x e.$$

41. If $f'(x) = \sqrt{2x^2 - 1}$ and $y = f(x^2)$ then $\left.\frac{dy}{dx}\right|_{x=1}$ is equal to

- (1) 2 (2) 1 (3) -2 (4) -1

Sol. Answer (1)

$$y = f(x^2) \Rightarrow \frac{dy}{dx} = f'(x^2) 2x = (2x) \sqrt{2x^4 - 1}$$

$$\text{At } x = 1, \frac{dy}{dx} = 2 \cdot 1 \cdot \sqrt{2-1}$$

42. Let the function $f(x)$ be defined as

$$f(x) = \begin{cases} \frac{\ln x - 1}{x - e}, & x \neq e \\ k, & x = e \end{cases}$$

The value of k , for which the function is continuous at $x = e$, is equal to

- (1) e (2) $\frac{1}{e}$ (3) e^2 (4) $-e$

Sol. Answer (2)

$$\lim_{x \rightarrow e} \frac{\ln x - 1}{x - e} = f'(e) = \frac{1}{e}$$

$$\therefore k = \frac{1}{e}$$

43. Rolle's theorem is not applicable to $f(x) = |x|$ in $[-2, 2]$ because

- (1) $f(x)$ is not continuous in $[-2, 2]$ (2) $f(x)$ is not derivable in $(-2, 2)$
 (3) $f(2) \neq f(-2)$ (4) It is applicable

Sol. Answer (2)

$f(x) = |x|$ is continuous in $[-2, 2]$

Also, $f(2) = f(-2) = 2$, but $f(x)$ is not derivable at $x = 0$

44. Lagrange's mean value theorem is not applicable to $f(x)$ in $[1, 4]$ where $f(x) =$

- (1) $x^2 - 2x$ (2) $|x - 2|$ (3) $x|x|$ (4) x^3

Sol. Answer (2)

$f(x) = |x - 2|$ is not derivable at $x = 2 \in (1, 4)$

Hence, LMVT is not applicable to $f(x) = |x - 2|$ in $[1, 4]$.

45. The value of c in Lagrange's mean value theorem for the function $f(x) = |x|$ in the interval $[-1, 1]$ is

- (1) 0 (2) $\frac{1}{2}$
 (3) $-\frac{1}{2}$ (4) Nonexistent in the interval

Sol. Answer (4)

By Lagrange's $\frac{f(1) - f(-1)}{1 - (-1)} = f'(c)$

i.e., $0 = f'(c)$ where $-1 < c < 1$

If $x > 0$, $f(x) = x$ therefore $f'(x) = 1$ and

If $x < 0$, $f(x) = -x$ therefore $f'(x) = -1$

Hence, there is no $c \neq 0$ for which $0 = f'(c)$ holds. Also at $x = 0$, $f(x)$ is not differentiable.

Hence, c has no value.

46. The value of c according to Rolle's theorem for the function $f(x) = x^3 - 9x^2 + 26x - 24$ in the interval $[2, 4]$ is

- (1) $9 + \sqrt{3}$ (2) $9 - \sqrt{3}$ (3) $\frac{9 \pm \sqrt{3}}{3}$ (4) $9 \pm \sqrt{3}$

Sol. Answer (3)

$f(x)$ is a polynomial

$\therefore$ It is continuous in $[2, 4]$ and is differentiable in $(2, 4)$ and $f(2) = 0$, $f(4) = 0$

$\therefore$ Rolle's theorem is applicable

$\therefore f'(c) = 0$ $c \in [2, 4]$

$f'(x) = 3x^2 - 18x + 26$

$3c^2 - 18c + 26 = 0$

$\therefore c = \frac{18 \pm \sqrt{12}}{6} = \frac{9 \pm \sqrt{3}}{3}$

47. The value of c according to Lagrange's mean value theorem for the function $f(x) = 2x^2 - 10x + 29$ in the interval $[2, 7]$ is

(1) 4.5 (2) 5 (3) 6 (4) 4

Sol. Answer (1)

$f(x)$ being a polynomial is continuous and differentiable in the given interval

$$\therefore f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$4c - 10 = \frac{40}{5} \Rightarrow c = \frac{9}{2} = 4.5$$

48. If $f(x) = x^\alpha \cdot \log x$ and $f(0) = 0$, then the value of α for which Rolle's theorem can be applied in $[0, 1]$ is

(1) -2 (2) -1 (3) 0 (4) $\frac{1}{2}$

Sol. Answer (4)

$f(x) = x^\alpha \log x$ cannot be defined for $\alpha = -2, -1$

for $\alpha = 0$ $f(x) = \log x$ $f(0) \neq 0$

$\therefore \alpha = \frac{1}{2}$ is only possibility

49. In $[0, 1]$, Lagrange mean value theorem is NOT applicable to

$$(1) f(x) = \begin{cases} \frac{1}{2} - x, & x < \frac{1}{2} \\ \left(\frac{1}{2} - x\right)^2, & x \geq \frac{1}{2} \end{cases}$$

$$(2) f(x) = \begin{cases} \frac{\sin x}{x}, & x \neq 0 \\ 1, & x = 0 \end{cases}$$

$$(3) f(x) = x|x|$$

$$(4) f(x) = |x|$$

Sol. Answer (1)

50. If f be a function of ' x ' such that $f(9) = 9$, $f'(9) = 3$, then $\lim_{x \rightarrow 9} \frac{\sqrt{f(x)} - 3}{\sqrt{x} - 3}$ is

(1) 9 (2) 3 (3) 1 (4) $\frac{1}{3}$

Sol. Answer (2)

$$\begin{aligned} \lim_{x \rightarrow 9} \frac{\sqrt{f(x)} - 3}{\sqrt{x} - 3} &= \lim_{x \rightarrow 9} \frac{f(x) - 3^2}{\sqrt{f(x)} + 3} \cdot \frac{\sqrt{x} + 3}{x - 3^2} \\ &= \lim_{x \rightarrow 9} \frac{f(x) - 9}{x - 9} \cdot \frac{\sqrt{x} + 3}{\sqrt{f(x)} + 3} \\ &= \lim_{x \rightarrow 9} \frac{f(x) - f(9)}{x - 9} \cdot \lim_{x \rightarrow 9} \frac{\sqrt{x} + 3}{\sqrt{f(x)} + 3} \\ &= f'(9) \frac{\sqrt{9} + 3}{\sqrt{f(9)} + 3} = 3 \left(\frac{3 + 3}{\sqrt{9} + 3} \right) = 3 \end{aligned}$$

51. If $f(x) = \begin{cases} \frac{1}{1+e^{1/x}}, & x \neq 0 \\ 0, & x = 0 \end{cases}$ then $f(x)$ is

- (1) Continuous at $x = 0$ (2) Continuous and differentiable at $x = 0$
 (3) Continuous but not differentiable at $x = 0$ (4) Discontinuous at $x = 0$

Sol. Answer (4)

$$f(0) = 0$$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{1}{1+e^{\frac{1}{x}}} = \lim_{x \rightarrow 0^+} \frac{1}{e^{\frac{1}{x}} \left(e^{-\frac{1}{x}} + 1 \right)} = \lim_{x \rightarrow 0^+} \frac{e^{-\frac{1}{x}}}{e^{-\frac{1}{x}} + 1} = \frac{0}{0+1} = 0$$

$$\text{and } \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \frac{1}{1+e^{\frac{1}{x}}} = \frac{1}{1+0} = 1$$

$$\text{Thus, } \lim_{x \rightarrow 0^+} f(x) \neq \lim_{x \rightarrow 0^-} f(x)$$

$\therefore f(x)$ is discontinuous at $x = 0$.

52. Let $f(x) = \sqrt{1-\sqrt{1-x^2}}$, then $f(x)$ is

- (1) Continuous on $[-1, 1]$ and differentiable on $(-1, 1)$
 (2) Continuous on $[-1, 1]$ and differentiable on $(-1, 0) \cup (0, 1)$
 (3) Continuous and differentiable on $(-1, 1)$
 (4) Discontinuous on $[-1, 1]$

Sol. Answer (2)

$f(x)$ is continuous in $[-1, 1]$

$$\begin{aligned} \text{But } f'(x) &= \frac{1}{2} \frac{1}{\sqrt{1-\sqrt{1-x^2}}} \cdot \frac{1}{2} \frac{1}{\sqrt{1-x^2}} (-2x) \\ &= \frac{-x}{2\sqrt{1-x^2} \sqrt{1-\sqrt{1-x^2}}} \end{aligned}$$

which does not exist when $\sqrt{1-x^2} = 0, 1$

$\Rightarrow f(x)$ is differentiable at all $x \in [-1, 1]$ except when $x^2 = 0, 1$.

$\Rightarrow f(x)$ is derivable in $(-1, 0) \cup (0, 1)$.

53. Domain of differentiability of the function $f(x) = |x-2| \cos x$ is

- (1) R (2) $R - \{2\}$ (3) $(0, \infty)$ (4) $\{2\}$

Sol. Answer (2)

As $|x-2|$ is not derivable when $x-2 = 0$, i.e. when $x = 2$, therefore $f(x) = |x-2| \cos x$ is derivable at all

$$x \in R \text{ except at } x = 2 \text{ and } f'(x) = |x-2| (-\sin x) + \frac{(\cos x)(x-2)}{|x-2|}.$$

54. Let $f(x) = \frac{\sin(\pi[x + \pi])}{1 + [x]^2}$, where $[]$ denotes the greatest integer function, then $f(x)$ is

- (1) Continuous and differentiable at all $x \in R$ (2) Continuous but not differentiable at some x
 (3) Differentiable but not continuous at $x = 0$ (4) Neither continuous nor differentiable at $x = 0$

Sol. Answer (1)

As $[x + \pi]$ is always an integer and sine of an integral multiple of π is zero, therefore, numerator of $f(x)$ is 0 for all $x \in R$ and denominator is always positive

$\Rightarrow f(x) = 0$ for all $x \in R$

So, $f(x)$ is a constant function with $D_f = R$

Hence, $f(x)$ is continuous and derivable at all $x \in R$.

55. If $f(x) = x^2 + \frac{x^2}{1+x^2} + \frac{x^2}{(1+x^2)^2} + \dots$ upto ∞ . Then

- (1) $\lim_{x \rightarrow 0} f(x)$ does not exist (2) $f(x)$ is continuous but not differentiable at $x = 0$
 (3) $f(x)$ is discontinuous at $x = 0$ (4) $f(x)$ is differentiable at $x = 0$

Sol. Answer (3)

$$f(0) = 0 + 0 + 0 + \dots = 0$$

$$\begin{aligned} \lim_{x \rightarrow 0} f(x) &= \lim_{x \rightarrow 0} \left\{ x^2 + \frac{x^2}{(1+x^2)} + \frac{x^2}{(1+x^2)^2} + \dots \right\} \\ &= \lim_{x \rightarrow 0} x^2 \left\{ 1 + \frac{1}{(1+x^2)} + \frac{1}{(1+x^2)^2} + \dots \right\} \\ &= \lim_{x \rightarrow 0} x^2 \left\{ \frac{1}{1 - \frac{1}{1+x^2}} \right\} \quad \left(\because x \rightarrow 0, \therefore x \neq 0 \Rightarrow 0 < \frac{1}{1+x^2} < 1 \right. \\ &\quad \left. \Rightarrow \text{Series inside the brackets is an infinite G.P. with } |C.R.| < 1 \right) \\ &= \lim_{x \rightarrow 0} (1+x^2) = 1 \end{aligned}$$

So, $f(0) \neq \lim_{x \rightarrow 0} f(x)$

56. Let $f(x) = \begin{cases} \frac{|x+1|}{\tan^{-1}(x+1)}, & x \neq -1 \\ 1, & x = -1 \end{cases}$. Then $f(x)$ is

- (1) Continuous at $x = -1$ (2) Differentiable at $x = -1$
 (3) Discontinuous at $x = -1$ (4) Continuous but not derivable at $x = -1$

Sol. Answer (3)

$$\begin{aligned} \lim_{x \rightarrow -1^+} f(x) &= \lim_{x \rightarrow -1^+} \frac{|x+1|}{\tan^{-1}|x+1|} \\ &= \lim_{x \rightarrow -1^+} \frac{x+1}{\tan^{-1}(x+1)} = 1 \quad (\because x \rightarrow -1^+ \Rightarrow x+1 \rightarrow 0^+ \text{ and } \lim_{t \rightarrow 0} \frac{\tan t}{t} = 1 = \lim_{s \rightarrow 0} \frac{s}{\tan^{-1} s}) \end{aligned}$$

$$\begin{aligned} \text{and } \lim_{x \rightarrow -1^+} f(x) &= \lim_{x \rightarrow -1^+} \frac{|x+1|}{\tan^{-1}(x+1)} \\ &= \lim_{x \rightarrow -1^+} \frac{-(x+1)}{\tan^{-1}(x+1)} = -1. \text{ Hence } f(x) \text{ is not continuous at } x = -1. \end{aligned}$$

57. The value of $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x-h)}{h}$ is equal to

- (1) $f'(x)$ (2) 0 (3) $2f'(x)$ (4) $-f'(x)$

Sol. Answer (3)

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f(x+h) - f(x-h)}{h} &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} - \lim_{h \rightarrow 0} \frac{f(x-h) - f(x)}{h} \\ &= f'(x) + f'(x) = 2f'(x). \end{aligned}$$

58. If $y = \frac{e^x + 1}{e^x - 1}$, then $\frac{y^2}{2} + \frac{dy}{dx}$ is equal to

- (1) 1 (2) -1 (3) $-\frac{1}{2}$ (4) $\frac{1}{2}$

Sol. Answer (4)

$$y = \frac{e^x + 1}{e^x - 1}$$

$$\frac{dy}{dx} = \frac{(e^x - 1)e^x - (e^x + 1)e^x}{(e^x - 1)^2}$$

$$= \frac{e^x(e^x - 1 - e^x - 1)}{(e^x - 1)^2} = -\frac{2e^x}{(e^x - 1)^2}$$

$$\therefore \frac{y^2}{2} + \frac{dy}{dx} = \frac{(e^x + 1)^2}{2(e^x - 1)^2} - \frac{2e^x}{(e^x - 1)^2}$$

$$= \frac{(e^x + 1)^2 - 4e^x}{2(e^x - 1)^2}$$

$$= \frac{(e^x - 1)^2}{2(e^x - 1)^2}$$

59. If $f(x) = e^x g(x)$, $g(0) = 2$, $g'(0) = 1$, then $f'(0) =$

- (1) 1 (2) 3 (3) 2 (4) 0

Sol. Answer (2)

$$f(x) = e^x g(x)$$

$$f'(x) = e^x g(x) + e^x g'(x)$$

$$f'(0) = e^0 g(0) + e^0 g'(0) = g(0) + g'(0)$$

$$= 2 + 1 = 3.$$

60. If $ax^2 + 2hxy + by^2 = 1$, then $\frac{d^2y}{dx^2}$ is equal to

(1) $\frac{h^2 - ab}{(hx + by)^2}$

(2) $\frac{h^2 - ab}{(hx + by)^3}$

(3) $\frac{ab - h^2}{(hx + by)^3}$

(4) $\frac{ab - h^2}{(hx + by)^2}$

Sol. Answer (2)

$$ax^2 + 2hxy + by^2 = 1$$

$$\Rightarrow 2ax + 2h\left(x\frac{dy}{dx} + y\right) + 2by\frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{ax + hy}{hx + by}$$

$$\frac{d^2y}{dx^2} = -\frac{(hx + by)\left(a + h\frac{dy}{dx}\right) - (ax + hy)\left(h + b\frac{dy}{dx}\right)}{(hx + by)^2}$$

$$= -\frac{(ahx + aby - ahx - h^2y) + (h^2x + hby - abx - hby)\frac{dy}{dx}}{(hx + by)^2}$$

$$= -\frac{y(ab - h^2) + x(h^2 - ab)\frac{dy}{dx}}{(hx + by)^2}$$

$$= -\frac{y(ab - h^2) - x(h^2 - ab)\left(\frac{ax + hy}{hx + by}\right)}{(hx + by)^3}$$

$$= -\frac{(hyx + by^2)(ab - h^2) - (h^2 - ab)(ax^2 + hxy)}{(hx + by)^3}$$

$$= -\frac{(ab - h^2)(hxy + by^2 + ax^2 + hxy)}{(hx + by)^3}$$

$$= \frac{h^2 - ab}{(hx + by)^3} \quad (\because ax^2 + 2hxy + by^2 = 1)$$

61. Derivative of the function $f(x) = \log_5(\log_7 x)$, where $x > 7$ is

(1) $\frac{1}{x(\log 5)(\log 7)(\log_7 x)}$

(2) $\frac{1}{x(\log 5)(\log 7)}$

(3) $\frac{1}{x(\log_e x)(\log_e 5)}$

(4) $\frac{1}{(\log 5)(\log 7)(x \log x)}$

Sol. Answer (3)

$$\text{Given, } y = \log_5 \log_7 x$$

$$y = \log_5 \frac{\log_e x}{\log_e 7}$$

$$= \log_5 \log_e x - \log_5 \log_e 7$$

$$y = \frac{\log_e \log_e x}{\log_e 5} - \log_5 \log_e 7$$

$$\therefore \frac{dy}{dx} = \frac{1}{\log_e 5} \left[\frac{1}{\log_e x} \cdot \frac{1}{x} \right]$$

62. Let $f(2) = 4$, $f'(2) = 4$. Then $\lim_{x \rightarrow 2} \frac{xf(2) - 2f(x)}{x - 2}$ is

(1) $-\frac{1}{3}$

(2) -2

(3) -4

(4) 3

Sol. Answer (3)

$$\lim_{x \rightarrow 2} \frac{xf(2) - 2f(x)}{x - 2}$$

Applying L' Hospital Rule

$$= \lim_{x \rightarrow 2} f(2) - 2f'(x)$$

$$= f(2) - 2f'(2)$$

$$= 4 - 2 \times 4$$

$$= -4$$

63. The value of $f(0)$ so that $f(x) = \frac{(4^x - 1)^3}{\sin \frac{x}{4} \cdot \log \left(1 + \frac{x^2}{3} \right)}$ is continuous everywhere is

(1) $3(\log 4)^3$

(2) $4(\log 4)^3$

(3) $12(\log 4)^3$

(4) $15(\log 4)^3$

Sol. Answer (3)

$$\begin{aligned} \lim_{x \rightarrow 0} f(x) &= \lim_{x \rightarrow 0} \frac{(4^x - 1)^3 x^3}{x^3} \cdot \frac{\frac{1}{\sin \frac{x}{4} \cdot \log \left(1 + \frac{x^2}{3} \right)} \cdot \frac{x^2}{3}}{\frac{x}{4} \cdot \frac{x^2}{3}} \\ &= \left(\frac{4^x - 1}{x} \right)^3 \cdot \frac{1}{\left(\frac{\sin x / 4}{\frac{x}{4}} \right)} \cdot \frac{1}{\frac{x}{4} \cdot \frac{x^2}{3} - \frac{x^4}{9} + \frac{x^6}{27} - \dots} \end{aligned}$$

$$\Rightarrow f(0) = 12(\log 4)^3$$

64. If $f(x) = \begin{cases} \frac{x^2 + 3x + p}{2(x^2 - 1)}, & x \neq 1 \\ \frac{5}{4}, & x = 1 \end{cases}$ is continuous at $x = 1$ then

(1) $p = 2$

(2) $p = 0$

(3) $p = -4$

(4) None of these

Sol. Answer (3)

$$\lim_{x \rightarrow 1} \frac{x^2 + 3x + p}{2(x^2 - 1)} = \frac{5}{4}$$

$$\Rightarrow \lim_{x \rightarrow 1} \frac{x^2 + 3x + p}{2(x+1)(x-1)} = \frac{5}{4}$$

For the existence of limit $(x-1)$ should be a factor of $x^2 + 3x + p$

$$\text{Hence } 1 + 3 + p = 0 \Rightarrow p = -4$$

65. The number of points at which the function $f(x) = \frac{1}{\log |x|}$ is discontinuous is

(1) 1

(2) 2

(3) 3

(4) 4

Sol. Answer (3)

$$f(x) = \frac{1}{\log |x|}$$

Clearly $f(x)$ is not defined at $x = 0, -1, 1$

Hence the given function is not continuous at 3 points ($x = 0, -1, 1$)

66. If $f(x) = \begin{cases} xe^{-\left(\frac{1}{|x|} + \frac{1}{x}\right)}, & x \neq 0 \\ 0, & x = 0 \end{cases}$, then $f(x)$ is

(1) Continuous for all x but not differentiable

(2) Neither differentiable nor continuous

(3) Discontinuous everywhere

(4) Continuous as well as differentiable for all x

Sol. Answer (1)

$$f(x) = \begin{cases} xe^{-\left(\frac{1}{|x|} + \frac{1}{x}\right)}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

$$f(x) = \begin{cases} x, & x < 0 \\ xe^{-2/x}, & x > 0 \\ 0, & x = 0 \end{cases}$$

$$\text{LHL} = 0$$

$$\text{RHL} = 0$$

$$f(0) = 0$$

$\therefore$ The function is continuous.

$$\text{LHD} = \lim_{h \rightarrow 0} \frac{h}{h} = 1$$

$$\text{RHD} = \lim_{h \rightarrow 0} \frac{he^{-2/h}}{h} = 0$$

$$\text{LHD} \neq \text{RHD}$$

Hence $f(x)$ is not differentiable.

67. The set of points where $f(x) = \frac{x}{1+|x|}$ is differentiable is

(1) $(-\infty, -1) \cup (-1, \infty)$

(2) $(-\infty, \infty)$

(3) $(0, \infty)$

(4) $(-\infty, 0) \cup (0, \infty)$

Sol. Answer (2)

$$f(x) = \frac{x}{1+|x|}$$

$$\Rightarrow f(x) = \begin{cases} \frac{x}{1-x}, & x < 0 \\ \frac{x}{1+x}, & x \geq 0 \end{cases}$$

$$\Rightarrow f'(x) = \begin{cases} \frac{1}{(1-x)^2}, & x < 0 \\ \frac{1}{(1+x)^2}, & x > 0 \end{cases}$$

Here we observe that

$$\text{LHD} = 1 = \text{L}f'(0)$$

$$\text{RHD} = 1 = \text{R}f'(0)$$

$\therefore$ Function is differentiable over $(-\infty, \infty)$

68. The function $|x^2 - 3x + 2| + \cos |x|$ is not differentiable at $x =$

(1) $-1, 2$

(2) $-1, -2$

(3) $1, 2$

(4) $-2, 1$

Sol. Answer (3)

$$f(x) = \begin{cases} x^2 - 3x + 2 + \cos x, & x < 0 \\ x^2 - 3x + 2 + \cos x, & 0 \leq x < 1 \\ -x^2 + 3x - 2 + \cos x, & 1 \leq x \leq 2 \\ x^2 - 3x + 2 + \cos x, & x \geq 2 \end{cases}$$

$$f'(x) = \begin{cases} 2x - 3 - \sin x, & x < 0 \\ 2x - 3 - \sin x, & 0 < x < 1 \\ -2x + 3 - \sin x, & 1 < x < 2 \\ 2x - 3 - \sin x, & x > 2 \end{cases}$$

At $x = 0$

$$\text{LHD} = -3$$

$$\text{RHD} = -3$$

$\Rightarrow (x)$ is differentiable at $x = 0$

At $x = 1$

$$\text{LHD} = -1 - \sin 1$$

$$\text{RHD} = 1 - \sin 1$$

$\Rightarrow f(x)$ is not differentiable at $x = 1$

At $x = 2$

$$\text{LHD} = -1 - \sin 2$$

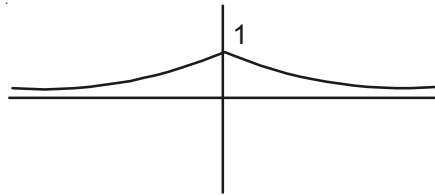
$$\text{RHD} = 1 - \sin 2$$

$\Rightarrow f(x)$ is not differentiable at $x = 2$

69. At $x = 0$, the function $y = e^{-|x|}$ is

- (1) Continuous (2) Continuous and differentiable
(3) Differentiable with derivative = 1 (4) Differentiable with derivative = -1

Sol. Answer (1)



From graph, $f(x)$ is continuous but not differentiable at $x = 0$.

70. Let $f(x) = \lambda + \mu|x| + \nu|x|^2$, where λ, μ, ν are real constants, then $f'(0)$ exists if

- (1) $\mu = 0$ (2) $\nu = 0$ (3) $\lambda = 0$ (4) $\mu = \nu$

Sol. Answer (1)

$$f(x) = \lambda + \mu|x| + \nu|x|^2$$

$$f(x) = \begin{cases} \lambda + \mu x + \nu x^2, & x \geq 0 \\ \lambda - \mu x + \nu x^2, & x < 0 \end{cases}$$

$$f'(x) = \begin{cases} \mu + 2\nu x, & x > 0 \\ -\mu + 2\nu x, & x < 0 \end{cases}$$

If $f'(0)$ exist then $\mu = 0$

71. Let $f(x) = [x]$, $g(x) = |x|$ and $f\{g(x)\} = h(x)$, where $[\cdot]$ is the greatest integer function. Then $h'(-1)$ is

- (1) 0 (2) $-\infty$ (3) Doesn't exist (4) None of these

Sol. Answer (3)

$$h(x) = f\{g(x)\} = [|x|]$$

$$h(x) = \begin{cases} [x], & x \geq 0 \\ [-x], & x < 0 \end{cases}$$

$h(x)$ is not differentiable at integral point

$\Rightarrow h'(-1)$ does not exist.

72. A function f is defined by $f(x^2) = x^3 \forall x > 0$ then $f'(4)$ equals

- (1) 1 (2) 2 (3) 3 (4) Not differentiable

Sol. Answer (3)

$$f(x^2) = x^3$$

Differentiating it w.r.t. x we get

$$2xf'(x^2) = 3x^2$$

$$\Rightarrow f'(x^2) = \frac{3x^2}{2x} = \frac{3x}{2}$$

$$\Rightarrow f'(4) = \frac{3 \times 2}{2} = 3$$

73. If $3\sin xy + 4\cos xy = 5$, then $\frac{dy}{dx}$ equals

(1) $\frac{y}{x}$

(2) $-\frac{y}{x}$

(3) $\frac{x}{y}$

(4) $-\frac{x}{y}$

Sol. Answer (2)

We have

$$3\sin xy + 4\cos xy = 5$$

$$\Rightarrow 3\cos xy \left[x \frac{dy}{dx} + y \right] - 4\sin xy \left[x \frac{dy}{dx} + y \right] = 0$$

$$\Rightarrow (3\cos xy - 4\sin xy) \left[x \frac{dy}{dx} + y \right] = 0$$

$$\Rightarrow \frac{xdy}{dx} = -y \Rightarrow \frac{dy}{dx} = -\frac{y}{x}$$

74. Let $f(x) = \max \{4, 1 + x^2, x^2 - 1\} \forall x \in \mathbb{R}$. Total number of points, where $f(x)$ is non-differentiable, is equal to

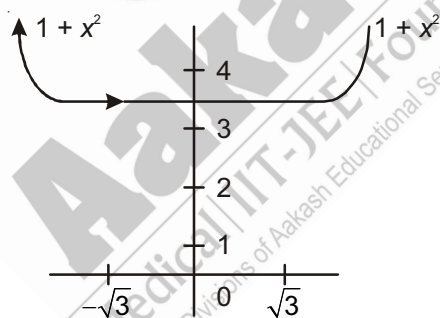
(1) 2

(2) 4

(3) 6

(4) 0

Sol. Answer (1)



From the graph of the given function, it follows that f is not differentiable at $x = \pm\sqrt{3}$.

75. Let $g(x)$ be the inverse of the function $f(x)$, and $f'(x) = \frac{1}{1+x^3}$ then $g'(x)$ equals

(1) $\frac{1}{(1+g^3)}$

(2) $\frac{1}{(1+f^3)}$

(3) $1 + g^3$

(4) $1 + f^3$

Sol. Answer (3)

$$f'(x) = \frac{1}{1+x^3}$$

$$f^{-1}(x) = g(x)$$

$$\therefore f(g(x)) = x$$

Differentiating it w.r.t. x we get

$$[f'(g(x))]g'(x) = 1$$

$$g'(x) = \frac{1}{f'(g(x))} = \frac{1}{\frac{1}{1+g^3}}$$

$$\Rightarrow g'(x) = 1 + (g(x))^3$$

76. If f is a real-valued differentiable function satisfying $|f(x) - f(y)| \leq (x - y)^2$, $x, y \in \mathbb{R}$ and $f(0) = 0$ then $f(1)$ equals

(1) 1

(2) 2

(3) 0

(4) -1

Sol. Answer (3)

$$|f(x) - f(y)| \leq (x - y)^2$$

$$\Rightarrow \lim_{x \rightarrow y} \frac{|f(x) - f(y)|}{|x - y|} \leq \lim_{x \rightarrow y} \frac{(x - y)^2}{|x - y|}$$

$$\Rightarrow |f'(y)| \leq 0 \Rightarrow |f'(y)| = 0$$

$$f'(y) = 0$$

$$f(y) = a$$

$$f(0) = a = 0 \text{ i.e., } f(y) = 0, \forall y \in \mathbb{R}$$

Hence $f(1) = 0$

77. $f(x) = \begin{cases} \frac{\sqrt{1+px} - \sqrt{1-px}}{x}; & -1 \leq x < 0 \\ \frac{2x+1}{x-2}; & 0 \leq x \leq 1 \end{cases}$; is continuous in the interval $[-1, 1]$, then p equals

(1) -1

(2) $-\frac{1}{2}$

(3) $\frac{1}{2}$

(4) 1

Sol. Answer (2)

$$f(x) = \begin{cases} \frac{\sqrt{1+px} - \sqrt{1-px}}{x}, & -1 \leq x < 0 \\ \frac{2x+1}{x-2}, & 0 \leq x \leq 1 \end{cases}$$

For the existence of limit at $x = 0$ we must have

$$\lim_{x \rightarrow 0^-} \frac{\sqrt{1+px} - \sqrt{1-px}}{x} = \lim_{x \rightarrow 0^+} \frac{2x+1}{x-2}$$

$$\Rightarrow \lim_{x \rightarrow 0^-} \frac{2p}{\sqrt{1+px} + \sqrt{1-px}} = -\frac{1}{2} \Rightarrow p = -\frac{1}{2}$$

78. If $f(x) = \begin{cases} \frac{a|x^2 - x - 2|}{2 + x - x^2}, & x < 2 \\ b, & x = 2 \\ \frac{x - [x]}{x - 2}, & x > 2 \end{cases}$

is continuous at $x = 2$ and $[\]$ is G.I.F., then values of a and b are

(1) $a = 1, b = 1$

(2) $a = 1, b = 2$

(3) $a = 2, b = 1$

(4) None of these

Sol. Answer (1)

$$f(x) = \begin{cases} \frac{a|x^2 - x - 2|}{2 + x - x^2}, & x < 2 \\ b, & x = 2 \\ \frac{x - [x]}{x - 2}, & x > 2 \end{cases}$$

$$f(x) = \begin{cases} a, & x < 2 \\ b, & x = 2 \\ 1, & x > 2 \end{cases}$$

At $x = 2$

$$\text{LHL} = a$$

$$\text{RHL} = 1$$

$$f(2) = b$$

For continuity $\text{LHL} = \text{RHL} = f(2)$

$$a = 1 = b.$$

79. The function $f(x) = \sin^{-1}(\cos x)$ is

(1) Continuous at $x = 0$

(2) Discontinuous at $x = 0$

(3) Differentiable at $x = 0$

(4) None of these

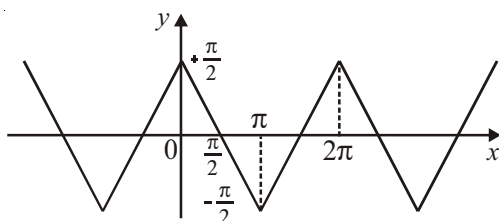
Sol. Answer (1)

We have

$$f(x) = \sin^{-1}(\cos x)$$

$$= \sin^{-1}\left(\sin\left(\frac{\pi}{2} - x\right)\right)$$

From graph it is clear $f(x)$ is continuous at $x = 0$ but not differentiable at $x = 0$



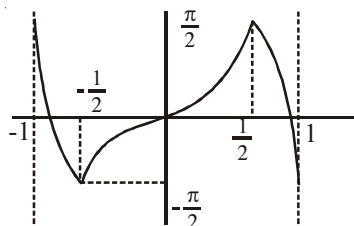
80. If $y = \sin^{-1}(3x - 4x^3)$, then the number of points in $[-1, 1]$, where y is not differentiable is

- (1) 0 (2) 1 (3) 2 (4) 3

Sol. Answer (3)

Graph of $\sin^{-1} 3x$

From graph its clear $f(x)$ is not differentiable at $x = -\frac{1}{2}$ and $\frac{1}{2}$.



81. Suppose $f(x)$ is differentiable at $x = 1$ and $\lim_{n \rightarrow 0} \frac{1}{n} f(1+n) = 5$, then $f'(1)$ equals

- (1) 6 (2) 5 (3) 4 (4) 3

Sol. Answer (2)

$$\lim_{n \rightarrow 0} \frac{1}{n} f(1+n) = 5$$

$$\Rightarrow \lim_{n \rightarrow 0} \frac{f(1+n)}{n} = 5$$

$$\Rightarrow \lim_{n \rightarrow 0} f'(1+n) = 5$$

$$\Rightarrow f'(1) = 5$$

82. Let $f(x+y) = f(x) \cdot f(y) \forall x, y \in R$. Suppose that $f(k) = 3, k \in R$ and $f'(0) = 11$, then $f'(k)$ is equal to

- (1) 3 (2) 11 (3) 33 (4) 0

Sol. Answer (3)

We have

$$f(x+y) = f(x) \cdot f(y) \quad x, y \in R$$

$$f(3) = 3, f(0) = 11$$

Let us put $x = 0, y = 0$ in $f(x+y) = f(x) \cdot f(y)$

$$\therefore f(0) = f(0)^2$$

$$\therefore f(0) = 1 \text{ as } f(0) = 0$$

$$\Rightarrow f(x) = 0, \forall x \in R \Rightarrow f(3) = 0, \neq 3, \text{ a contradiction}$$

$$\therefore f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{f(x) \cdot f(h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(x)[f(h) - 1]}{h} = \lim_{h \rightarrow 0} \frac{f(x)[f(h) - f(0)]}{h} = f(x) f'(0)$$

$$\Rightarrow f'(x) = f(x) f'(0)$$

$$\Rightarrow f'(3) = f(3) f'(0) = 3 \times 11 = 33$$

$$83. \text{ Let } f(x) = \begin{cases} x + a\sqrt{2} \sin x & ; 0 \leq x < \frac{\pi}{4} \\ 2x \cot x + b & ; \frac{\pi}{4} \leq x \leq \frac{\pi}{2} \\ a \cos 2x - b \sin x & ; \frac{\pi}{2} < x \leq \pi \end{cases}$$

continuous function $\forall x \in [0, \pi]$ then the value of $5\left(\frac{a}{b}\right)^2$ is

(1) 20

(2) 10

(3) 5

(4) 40

Sol. Answer (1)

$$f(x) = \begin{cases} x + a\sqrt{2} \sin x; & 0 \leq x < \frac{\pi}{4} \\ 2x \cot x + b; & \frac{\pi}{4} \leq x \leq \frac{\pi}{2} \\ a \cos 2x - b \sin x; & \frac{\pi}{2} < x \leq \pi \end{cases}$$

at $x = \frac{\pi}{4}$

$$\lim_{x \rightarrow \frac{\pi}{4}^-} f(x) = \lim_{x \rightarrow \frac{\pi}{4}^-} (x + a\sqrt{2} \sin x) = \frac{\pi}{4} + a, \text{ and } \lim_{x \rightarrow \frac{\pi}{4}^+} f(x) = \lim_{x \rightarrow \frac{\pi}{4}^+} 2x \cot x + b, \quad f\left(\frac{\pi}{4}\right) = \frac{\pi}{2} + b$$

Since $f(x)$ is continuous at $x = \frac{\pi}{4}$

$$\text{Hence } \lim_{x \rightarrow \frac{\pi}{4}^-} f(x) = \lim_{x \rightarrow \frac{\pi}{4}^+} f(x) = f\left(\frac{\pi}{4}\right)$$

$$\Rightarrow \frac{\pi}{4} + a = \frac{\pi}{2} + b \Rightarrow a - b = \frac{\pi}{4} \quad \dots(1)$$

At $x = \frac{\pi}{2}$ we observe that

$$\lim_{x \rightarrow \frac{\pi}{2}^-} f(x) = b, \quad \lim_{x \rightarrow \frac{\pi}{2}^+} f(x) = -a - b \text{ and } f\left(\frac{\pi}{2}\right) = b$$

Since $f(x)$ is continuous at $x = \frac{\pi}{2}$, hence $\lim_{x \rightarrow \frac{\pi}{2}^-} f(x) = \lim_{x \rightarrow \frac{\pi}{2}^+} f(x) = f\left(\frac{\pi}{2}\right)$

$$\Rightarrow b = -a - b = b$$

$$\Rightarrow 2b + a = 0 \quad \dots(2)$$

On solving (1) and (2), we get

$$-3b = \frac{\pi}{4}$$

$$\Rightarrow b = -\frac{\pi}{12} \text{ and } a = \frac{\pi}{6}$$

$$\text{Hence } 5\left(\frac{a}{b}\right)^2 = 20$$

84. The value of $f(1)$ so that the function $f(x) = \frac{9(x^{\frac{2}{3}} - 2x^{\frac{1}{3}} + 1)}{(x-1)^2}$, $x \neq 1$ is continuous at $x = 1$ is

(1) 1

(2) 2

(3) 3

(4) 4

Sol. Answer (1)

In order that $f(x)$ is continuous at $x = 1$, we must have

$$\lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} \frac{9(x^{2/3} - 2x^{1/3} + 1)}{(x-1)^2} = f(1)$$

$$\Rightarrow \lim_{x \rightarrow 1} \frac{9(x^{1/3} - 1)^2}{(x^{1/3} - 1)^2(x^{2/3} + 1 + x^{1/3})^2} = f(1)$$

$$\Rightarrow f(1) = 1.$$

85. Let $f(x)$ be a continuous function defined for $\forall x \in R$. If $f(x)$ take rational values $\forall x \in R$ and $f(2) = 198$, then $f(2^2) = \dots\dots$

(1) 99

(2) 198

(3) 33

(4) 4

Sol. Answer (2)

As $f(x)$ is continuous function in $[1, 3]$ and takes rational value only hence $f(x)$ is a constant function.

$$\therefore f(x) = f(2) = 198$$

$$\Rightarrow f(2^2) = 198$$

86. Let $F(x) = f(x).g(x).h(x) \forall x \in R$ and $f(x), g(x), h(x)$ are differentiable functions. At some point $x_0 \in R$.

$$F'(x_0) = 21 F(x_0), f'(x_0) = 4f(x_0)$$

$$g'(x_0) = -7g(x_0) \text{ and } h'(x_0) = kh(x_0).$$

Then $3k + 28$ is equal to

(1) 20

(2) 50

(3) 100

(4) 200

Sol. Answer (3)

$$F(x) = f(x).g(x).h(x)$$

$$\Rightarrow F'(x) = f'(x).g(x).h(x) + f(x).g'(x).h(x) + f(x).g(x).h'(x)$$

$$\Rightarrow F'(x_0) = f'(x_0).g(x_0).h(x_0) + f(x_0).g'(x_0).h(x_0) + f(x_0).g(x_0).h'(x_0)$$

$$\Rightarrow 21F(x_0) = 4f(x_0).g(x_0).h(x_0) - 7f(x_0).g(x_0).h(x_0) + kf(x_0).g(x_0).h(x_0)$$

$$\Rightarrow 21F(x_0) = (-3 + k)F(x_0), k = 24$$

$$\text{Hence } 3k + 28 = 100.$$

87. Let $f(x)$ is a polynomial satisfying $f(x).f(y) = f(x) + f(y) + f(xy) - 2 \forall x, y$ and $f(2) = 1025$, then the value of $\lim_{x \rightarrow 2} f'(x)$ is

- (1) $5 \cdot 2^8$ (2) $5 \cdot 2^9$ (3) $5 \cdot 2^7$ (4) $5 \cdot 2^{10}$

Sol. Answer (4)

$$\text{Given } f(x).f(y) = f(x) + f(y) + f(xy) - 2$$

$$\text{Put } y = \frac{1}{x}$$

$$\Rightarrow f(x) \cdot f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right) + f(1) - 2$$

$$\text{Put } x = 1$$

$$\Rightarrow \{f(1)\}^2 = 3f(1) - 2$$

$$\Rightarrow \{f(1)\}^2 - 3f(1) + 2 = 0$$

$$\Rightarrow (f(1) - 1)(f(1) - 2) = 0$$

$$\Rightarrow f(1) = 1, 2$$

$$\Rightarrow f(1) = 2, (\because f(1) \neq 1)$$

$$\therefore f(x) \cdot f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right) + 2 - 2$$

$$\Rightarrow f(x) \cdot f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$$

$$\Rightarrow f(x) = 1 \pm x^n$$

$$\therefore f(2) = 1 \pm 2^n = 1025$$

$$\Rightarrow 2^n = 1024 = 2^{10}$$

$$\Rightarrow n = 10$$

$$\therefore f(x) = 1 + x^{10}$$

$$\Rightarrow f'(x) = 10x^9$$

$$\therefore \lim_{x \rightarrow 2} f'(x) = \lim_{x \rightarrow 2} 10x^9 = 10 \times 2^9 = 5 \times 2^{10}$$

88. If $y = \frac{1}{1+x^{a-b}+x^{c-b}} + \frac{1}{1+x^{b-c}+x^{a-c}} + \frac{1}{1+x^{b-a}+x^{c-a}}$ then find $\frac{dy}{dx}$ at e^{abc} .

- (1) 0 (2) 1 (3) 2 (4) 3

Sol. Answer (1)

$$y = \frac{1}{1+x^{a-b}+x^{c-b}} + \frac{1}{1+x^{b-c}+x^{a-c}} + \dots$$

$$y = \frac{x^b}{x^a+x^b+x^c} + \frac{x^c}{x^a+x^b+x^c} + \frac{x^a}{x^a+x^b+x^c}$$

$$y = \frac{x^a+x^b+x^c}{x^a+x^b+x^c} = 1$$

$$\frac{dy}{dx} = 0$$

89. Let $f(x) = \begin{cases} x^p \sin\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$ and A and B are two sets such that A and B are continuous and differential of $f(x)$ at $x = 0$ respectively then find $A \cap B$.

- (1) $(1, \infty)$ (2) $(2, \infty)$ (3) $(-\infty, 1)$ (4) $(-\infty, 0)$

Sol. Answer (1)

$$A = \lim_{x \rightarrow 0} x^p \sin\left(\frac{1}{x}\right)$$

$$= 0, \text{ when } p > 0$$

$$\Rightarrow A = (0, \infty)$$

$$\therefore B = \lim_{x \rightarrow 0} \frac{x^p \sin\left(\frac{1}{x}\right) - 0}{x - 0}$$

$$= \lim_{x \rightarrow 0} x^{p-1} \sin\left(\frac{1}{x}\right)$$

$$= 0, \text{ when } p - 1 > 0$$

$$\Rightarrow B = (1, \infty)$$

$$\therefore A \cap B = \{(0, \infty) \cap (1, \infty)\} = (1, \infty)$$

90. If $f(x) = \begin{cases} \sin x & x < 0 \\ \cos x - |x - 1| & x \geq 0 \end{cases}$ then $g(x) = f(|x|)$ is non-differentiable for

- (1) No value of x (2) Exactly three values of x (3) Exactly one value of x (4) None of these

Sol. Answer (3)

$$f(|x|) = \cos x - ||x| - 1|, x \in \mathbb{R} \text{ [as } |x| < 0 \text{ is not possible \& } |x| \geq 0 \text{ is true } \forall x \in \mathbb{R}]$$

$$f(|x|) = \cos x - ||x| - 1|, x \in \mathbb{R}$$

Which is non-differentiable at $x = 0$ and when $|x| - 1 = 0$ or $x \pm 1$

Hence $f(|x|)$ has exactly 3 points of non-differentiability.

SECTION - B

Objective Type Questions (More than one options are correct)

1. Given the function $f(x) = \frac{1}{1-x}$. The points of discontinuity of the composite function $f[f(f(x))]$ are given by
- (1) 0 (2) 1 (3) 2 (4) -1

Sol. Answer (1, 2)

$$f(x) = \frac{1}{1-x}$$

$f(x)$ discontinuous at $x = 1$

$$f(f(x)) = \frac{1}{1 - \frac{1}{1-x}} = \frac{1-x}{-x} = 1 - \frac{1}{x} = \frac{x-1}{x}$$

$f(f(x))$ discontinuous at $x = 0$

$$f(f(f(x))) = \frac{1}{1 - \frac{x-1}{x}} = \frac{x}{x-x+1} = x$$

Although $f(f(f(x))) = x$ but $f(f(f(x)))$ exist only when $f(f(x))$ and $f(x)$ exist.

Hence $f(f(f(x)))$ is discontinuous at $x = 0$ and $x = 1$

2. A function is defined as follows

$$f(x) = \begin{cases} x^3 & ; x^2 < 1 \\ x & ; x^2 \geq 1 \end{cases} \text{ then function is}$$

(1) Continuous at $x = 1$

(2) Differentiable at $x = 1$

(3) Continuous but not differentiable at $x = 1$

(4) None of these

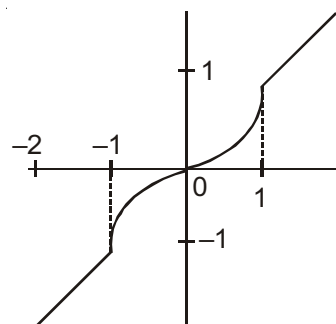
Sol. Answer (1, 3)

We have

$$f(x) = \begin{cases} x^3, & x^2 < 1 \\ x, & x^2 \geq 1 \end{cases}$$

$$= \begin{cases} x^3, & -1 < x < 1 \\ x, & x \geq 1 \text{ or } x \leq -1 \end{cases}$$

From graph it is clear that $f(x)$ is continuous at $x = 1$ but not differentiable at $x = 1$.



3. If $y = \tan x \tan 2x \tan 3x$, then $\frac{dy}{dx}$ equals

(1) $3\sec^2 3x \tan x \tan 2x + 2\sec^2 2x \tan 3x \tan x + \sec^2 x \tan 2x \tan 3x$

(2) $2y(\operatorname{cosec} 2x + 2 \operatorname{cosec} 4x + 3 \operatorname{cosec} 6x)$

(3) $3\sec^2 3x - 2\sec^2 2x - \sec^2 x$

(4) $\sec^2 x + 2\sec^2 2x + 3\sec^2 3x$

Sol. Answer (1, 2, 3)

$$y = \tan x \tan 2x \tan 3x$$

Differentiating

$$\frac{dy}{dx} = \tan x \tan 2x \frac{d \tan 3x}{dx} + \tan 2x \tan 3x \frac{d \tan x}{dx} + \tan x \tan 3x \frac{d \tan 2x}{dx}$$

$$\frac{dy}{dx} = \tan x \tan 2x \sec^2 3x \cdot 3 + \tan 2x \tan 3x \sec^2 x + \tan x \tan 3x \sec^2 2x \cdot 2$$

$$= \tan 2x \tan 3x \sec^2 x + 2 \tan x \tan 3x \sec^2 2x + 3 \tan x \tan 2x \sec^2 3x$$

Again $y = \tan x \tan 2x \tan 3x$

$$y = \tan 3x - \tan 2x - \tan x$$

Differentiating

$$\frac{dy}{dx} = \sec^2 3x \cdot 3 - \sec^2 2x \cdot 2 - \sec^2 x = 3\sec^2 3x - 2\sec^2 2x - \sec^2 x$$

Again

$$y = \tan x \tan 2x \tan 3x$$

$$\Rightarrow \log y = \log \tan x + \log \tan 2x + \log \tan 3x$$

$$\Rightarrow \frac{dy}{dx} = 2y [\operatorname{cosec} 2x + 2\operatorname{cosec} 4x + 3\operatorname{cosec} 6x]$$

4. Which of the following function(s) not defined at $x = 0$ has / have removable discontinuity at $x = 0$?

(1) $\frac{1}{(1+2^{\cot x})}$

(2) $\cos\left(\frac{|\sin x|}{x}\right)$

(3) $x \sin \frac{\pi}{x}$

(4) $\frac{1}{\ln |x|}$

Sol. Answer (2, 3, 4)

We observe that $\frac{|\sin x|}{x} \rightarrow 1$ for $x \rightarrow 0^+$ and $\frac{|\sin x|}{x} \rightarrow -1$, for $x \rightarrow 0^-$

But in either case $\lim_{x \rightarrow 0} \cos\left(\frac{|\sin x|}{x}\right) = \cos 1$

Also $x \sin \frac{\pi}{x} \rightarrow 0$ as $x \rightarrow 0$ and $\frac{1}{\ln |x|} \rightarrow 0$ as $x \rightarrow 0$

Hence the functions $\frac{|\sin x|}{x}$, $x \sin \frac{\pi}{x}$, $\frac{1}{\ln |x|}$ have removable discontinuity while $f(x) = \frac{1}{1+2^{\cot x}}$ has non-removable discontinuity as L.H.L. $\neq$ R.H.L.

5. Let $f(x) = \lim_{x \rightarrow 0} \frac{x^{2n} \sin^n x}{x^{2n} - \sin^{2n} x}$, $x \neq 0$ and $f(0) = L$ for some n . If $f(x)$ is continuous at $x = 0$, then

- (1) $n = 3, L = 6$ (2) $n = 2, L = \frac{3}{2}$
 (3) $n = 3, L = 0$ (4) $n = 2, L = 6$

Sol. Answer (2, 3)

$$\lim_{x \rightarrow 0} \frac{x^{2n} \sin^n x}{x^{2n} - \sin^{2n} x} = \lim_{x \rightarrow 0} \frac{x^{2n} \cdot \sin^n x}{(x^n - \sin^n x)(x^n + \sin^n x)}$$

$$\lim_{x \rightarrow 0} \frac{x^{2n} \sin^n x}{(x - \sin x)(x^{n-1} + x^{n-2} \sin x + \dots + \sin^{n-1} x)(x^n + \sin^n x)}$$

$$L = \lim_{x \rightarrow 0} \frac{\frac{\sin^n x}{x^n} \cdot x^{2n} \cdot x^{-n+2}}{\left(\frac{x - \sin x}{x^3}\right) \left(\frac{x^{n-1} + x^{n-2} \sin x + \sin^{n-1} x}{x^{n-1}}\right) \left(\frac{x^n + \sin^n x}{x^n}\right)}$$

if $n = 2$

$$L = \frac{1}{\frac{1}{6} \times 2 \times 2} = \frac{3}{2}$$

$$2n - n - 2 = 0, n = 2$$

i.e., $n = 2$ and $L = \frac{3}{2}$ and if $n = 3$ then $L = 0$

6. Let $f(x) = \begin{cases} 1 - \sqrt{1 - x^2} & \text{for } -1 \leq x \leq 1 \\ 1 + \log \frac{1}{x} & \text{for } x > 1 \end{cases}$, then

- (1) f is continuous at $x = 1$ (2) f is not differentiable at $x = 1$
 (3) f is continuous and differentiable at $x = 1$ (4) $f'(x)$ exists for all $x \in (0, 1)$

Sol. Answer (1, 2, 4)

We have

$$f(1-0) = 1 - \sqrt{1-1} = 1$$

$$f(1+0) = 1 + \log 1 = 1$$

Therefore f is continuous at $x = 1$

$$\text{Now } f'(x) = \frac{x}{\sqrt{1-x^2}}$$

for $0 \leq x < 1$ and at $x = 1$, $f'(1-0)$ does not exist

Hence f is not differentiable at $x = 1$

$f'(x)$ exists for all $x \in (0, 1)$ then (1), (2) and (4) are true.

7. A function is defined as $f(x) = \begin{cases} 5^x, & x \leq 0 \\ |x-1|, & x \geq 0 \end{cases}$. Then $f(x)$ is

- (1) Continuous at $x = 0$ (2) Continuous at $x = 1$
 (3) Differentiable at $x = 0$ (4) Differentiable at $x = 1$

Sol. Answer(1, 2)

$$f(x) = \begin{cases} 5^x & x \leq 0 \\ 1-x & 0 < x < 1 \\ x-1 & x \geq 1 \end{cases}$$

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{h \rightarrow 0} 5^{(0-h)} = 1$$

$$\lim_{x \rightarrow 0^+} f(x) = 1$$

$$\text{and } \lim_{x \rightarrow 1^-} f(x) = 0, \lim_{x \rightarrow 1^+} f(x) = 0$$

Hence, function $f(x)$ is continuous at $x = 0$ and 1 .

$$f'(0^-) = \log_e 5, \text{ and } f'(0^+) = -1$$

Hence, $f(x)$ is not differentiable at $x = 0$ and $x = 1$

8. A differentiable function $f: R^+ \rightarrow R$ satisfies $f(xy) = f(x) + f(y) \forall x, y \in R^+$. If $f(16) = 3$. Then the value of $f(2)$ is greater than

- (1) $\frac{7}{8}$ (2) $\frac{9}{8}$
 (3) $\frac{5}{16}$ (4) $\frac{3}{8}$

Sol. Answer(3, 4)

$$f(xy) = f(x) + f(y) \forall x, y \in R^+$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{f\left(x\left(1+\frac{h}{x}\right)\right) - f(x)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x) + f\left(1+\frac{h}{x}\right) - f(x)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f\left(1+\frac{h}{x}\right) - f(1)}{x\left(\frac{h}{x}\right)}; \quad \because f(1) = 0$$

$$\therefore f'(x) = \frac{1}{x} f'(1)$$

$$\Rightarrow f(x) = f'(1) \ln x + \ln c$$

$$\text{if } x = 1 \text{ then } \log c = 0$$

$$\therefore f(x) = f'(1) \ln x$$

$$\therefore f(16) = f'(1) \ln 16 = 4f'(1) \ln 2$$

$$f'(1) = \frac{3}{4 \ln 2}$$

$$f(2) = f'(1) \cdot \ln 2 = \frac{3}{4}$$

9. Let $f : [a, b] \rightarrow [1, \infty)$ be a continuous function and let $g : \mathbb{R} \rightarrow \mathbb{R}$ be defined as

$$g(x) = \begin{cases} 0 & \text{if } x < a, \\ \int_a^x f(t) dt & \text{if } a \leq x \leq b, \text{ then} \\ \int_a^b f(t) dt & \text{if } x > b \end{cases}$$

[JEE(Advanced)-2014]

(1) $g(x)$ is continuous but not differentiable at a

(2) $g(x)$ is differentiable on $\mathbb{R}$

(3) $g(x)$ is continuous but not differentiable at b

(4) $g(x)$ is continuous and differentiable at either a or b but not both

Sol. Answer (1, 3)

$$g(a^-) = 0, g(a) = \int_a^a f(t) dt = 0, g(a^+) = \lim_{h \rightarrow 0} \int_a^{a+h} f(t) dt = 0$$

$$g(b^-) = \lim_{h \rightarrow 0} \int_a^{b-h} f(t) dt = \int_a^b f(t) dt$$

$$g(b) = \int_a^b f(t) dt = g(b^+)$$

Hence $g(x)$ is continuous at $x = a$ as well as $x = b$

$$\text{Now, } g'(a^-) = \lim_{h \rightarrow 0} \frac{g(a-h) - g(a)}{-h} = 0$$

$$g'(a^+) = \lim_{h \rightarrow 0} \frac{g(a+h) - g(a)}{h} = \lim_{h \rightarrow 0} \frac{\int_a^{a+h} f(t) dt - 0}{h} = \lim_{h \rightarrow 0} \frac{f(a+h)}{1} = f(a) (\neq 0)$$

Hence $g(x)$ is not differentiable at $x = a$.

$$g'(b^-) = \lim_{h \rightarrow 0} \frac{g(b-h) - g(b)}{-h} = \lim_{h \rightarrow 0} \frac{\int_a^{b-h} f(t) dt - \int_a^b f(t) dt}{-h} = \lim_{h \rightarrow 0} \frac{f(b-h)}{1} = f(b) (\neq 0)$$

$$g'(b^+) = \lim_{h \rightarrow 0} \frac{g(b-h) - g(b)}{-h} = \lim_{h \rightarrow 0} \frac{\int_a^{b-h} f(t) dt - \int_a^b f(t) dt}{-h} = \lim_{h \rightarrow 0} \frac{f(b-h)}{1} = f(b) (\neq 0)$$

Hence $g(x)$ is not differentiable at $x = b$.

10. Consider the hyperbola $H: x^2 - y^2 = 1$ and a circle S with center $N(x_2, 0)$. Suppose that H and S touch each other at a point $P(x_1, y_1)$ with $x_1 > 1$ and $y_1 > 0$. The common tangent to H and S at P intersects the x -axis at point M . If (l, m) is the centroid of the triangle $\triangle PMN$, then the correct expression(s) is(are)

[JEE(Advanced)-2015]

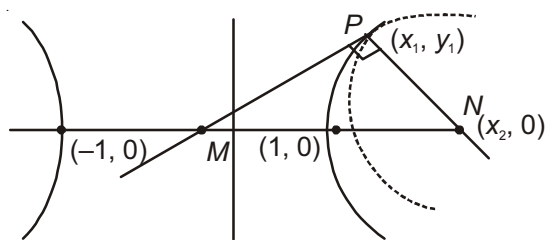
$$(1) \frac{dl}{dx_1} = 1 - \frac{1}{3x_1^2} \text{ for } x_1 > 1$$

$$(2) \frac{dm}{dx_1} = \frac{x_1}{3(\sqrt{x_1^2 - 1})} \text{ for } x_1 > 1$$

$$(3) \frac{dl}{dx_1} = 1 + \frac{1}{3x_1^2} \text{ for } x_1 > 1$$

$$(4) \frac{dm}{dy_1} = \frac{1}{3} \text{ for } y_1 > 0$$

Sol. Answer (1, 2, 4)

Equation of tangent at P on hyperbola.

$$xx_1 - yy_1 = 1$$

$$\text{Point } M\left(\frac{1}{x_1}, 0\right)$$

Equation of normal at P

$$\frac{x}{x_1} + \frac{y}{y_1} = 2$$

since $(x_2, 0)$ satisfies its

$$x_2 = 2x_1$$

$$\text{Centroid } (l, m) \equiv \left(x_1 + \frac{1}{3x_1}, \frac{y_1}{3}\right)$$

$$\frac{dl}{dx_1} = 1 - \frac{1}{3x_1^2}$$

$$\frac{dm}{dy_1} = \frac{1}{3}$$

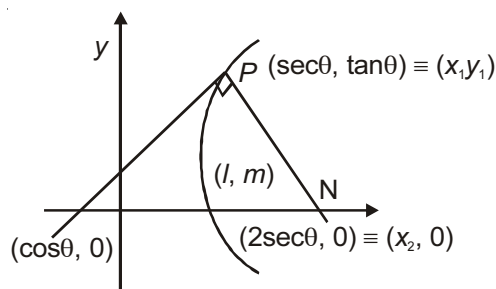
$$x_1^2 - y_1^2 = 1$$

$$y_1 = \sqrt{x_1^2 - 1}$$

$$m = \frac{\sqrt{x_1^2 - 1}}{3}$$

$$\frac{dm}{dx_1} = \frac{x_1}{3\sqrt{x_1^2 - 1}}$$

Alternative Method



$$l = \frac{3 \sec \theta + \cos \theta}{3}$$

$$m = \frac{\tan \theta}{3} = \frac{y_1}{3}$$

$$\frac{dl}{dx_1} = \frac{3 \sec \theta \tan \theta - \sin \theta}{3 \sec \theta \tan \theta} = 1 - \frac{1}{3 \sec^2 \theta} = 1 - \frac{1}{3x_1^2}$$

$$\frac{dm}{dx_1} = \frac{\frac{\sec^2 \theta}{3}}{\sec \theta \tan \theta} = \frac{\operatorname{cosec} \theta}{3} = \frac{x_1}{3\sqrt{x_1^2 - 1}}$$

$$\frac{dm}{dy_1} = \frac{1}{3}$$

11. Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function with $g(0) = 0$, $g'(0) = 0$ and $g'(1) \neq 0$. Let $f(x) = \begin{cases} \frac{x}{|x|} g(x), & x \neq 0 \\ 0, & x = 0 \end{cases}$

and $h(x) = e^{|x|}$ for all $x \in \mathbb{R}$. Let $(f \circ h)(x)$ denote $f(h(x))$ and $(h \circ f)(x)$ denote $h(f(x))$. Then which of the following is (are) true?

[JEE(Advanced)-2015]

- (1) f is differentiable at $x = 0$ (2) h is differentiable at $x = 0$
 (3) $f \circ h$ is differentiable at $x = 0$ (4) $h \circ f$ is differentiable at $x = 0$

Sol. Answer (1, 4)

$$f(x) = \begin{cases} g(x) & x > 0 \\ 0 & x = 0 \\ -g(x) & x < 0 \end{cases}$$

$$\text{LHD : } \lim_{x \rightarrow 0^-} (-g'(x)) = - \lim_{x \rightarrow 0^-} g'(x) = -0 = 0$$

$$\text{RHD : } \lim_{x \rightarrow 0^+} g'(x) = g'(0^+) = 0$$

Hence $f(x)$ is differentiable at $x = 0$

$$h(x) = e^{|x|} = \begin{cases} e^x & x \geq 0 \\ e^{-x} & x < 0 \end{cases}$$

$$\text{LHD : } \lim_{x \rightarrow 0^-} (-e^{-x}) = -1$$

$$\text{RHD : } \lim_{x \rightarrow 0^+} (e^x) = 1$$

Hence $h(x)$ is not differentiable at $x = 0$

$$f(h(x)) = \begin{cases} \frac{e^{|x|}}{|e^{|x|}|} g(e^{|x|}) \end{cases}$$

$$f(h(x)) = \begin{cases} g(e^x) & x \geq 0 \\ g(e^{-x}) & x < 0 \end{cases}$$

$$\text{LHD : } \lim_{x \rightarrow 0^-} g'(e^{-x}) \times -e^{-x} = g'(1) \times -1 = -g'(1)$$

$$\text{RHD : } \lim_{x \rightarrow 0^+} g'(e^x) \times e^x = g'(1)$$

Hence $f \circ h$ is non-differentiable at $x = 0$

$$h(f(x)) = \begin{cases} 1 & x = 0 \\ e^{\left| \frac{x}{|x|} g(x) \right|} & x \neq 0 \end{cases} = \begin{cases} 1 & x = 0 \\ e^{|g(x)|} & x \neq 0 \end{cases}$$

$$\text{LHD : } \lim_{x \rightarrow 0^-} e^{|g(x)|} \times \pm g'(x) = e^{|g(0^-)|} \times \pm (g'(0^-)) = 0$$

$$\text{RHD : } \lim_{x \rightarrow 0^+} e^{|g(x)|} \times \pm g'(x) = e^{|g(0^+)|} \times \pm (g'(0^+)) = 0$$

12. Let $f : \mathbb{R} \rightarrow \mathbb{R}$, $g : \mathbb{R} \rightarrow \mathbb{R}$ and $h : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable functions such that $f(x) = x^3 + 3x + 2$, $g(f(x)) = x$ and $h(g(g(x))) = x$ for all $x \in \mathbb{R}$. Then **[JEE(Advanced)-2016]**

$$(1) \quad g'(2) = \frac{1}{15}$$

$$(2) \quad h'(1) = 666$$

$$(3) \quad h(0) = 16$$

$$(4) \quad h(g(3)) = 36$$

Sol. Answer (2, 3)

$$g'(f(x)) \cdot f'(x) = 1$$

$$g'(2) \cdot f'(0) = 1$$

$$g'(2) = \frac{1}{f'(0)}$$

$$f'(x) = 3x^2 + 3$$

$$g'(2) = \frac{1}{3}$$

$$h(g(g(x))) = x$$

$$h(g(g(f(x)))) = f(x)$$

$$h(g(x)) = f(x)$$

$$h(g(3)) = f(3) = 38$$

$$h(g(f(x))) = f(f(x))$$

$$h(x) = f(f(x))$$

$$h'(x) = f'(f(x)) \cdot f'(x)$$

$$h'(1) = f'(f(1)) \cdot f'(1) = 111 \times 6 = 666$$

$$h(0) = f(f(0)) = f(2) = 16$$

13. Let $a, b \in \mathbb{R}$ and $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = a \cos(|x^3 - x|) + b|x| \sin(|x^3 + x|)$. Then f is

[JEE(Advanced)-2016]

(1) Differentiable at $x = 0$ if $a = 0$ and $b = 1$

(2) Differentiable at $x = 1$ if $a = 1$ and $b = 0$

(3) **NOT** differentiable at $x = 0$ if $a = 1$ and $b = 0$

(4) **NOT** differentiable at $x = 1$ if $a = 1$ and $b = 1$

Sol. Answer (1, 2)

$$f(x) = a \cos(|x^3 - x|) + b|x| \sin(|x^3 + x|)$$

$$\Rightarrow f(x) = a \cos(x^3 - x) + bx \sin(x^3 + x) \text{ for } x \in \mathbb{R}$$

So $f(x)$ is differentiable for all $x \in \mathbb{R}$

14. Let $f: \left[-\frac{1}{2}, 2\right] \rightarrow \mathbb{R}$ and $g: \left[-\frac{1}{2}, 2\right] \rightarrow \mathbb{R}$ be functions defined by $f(x) = \lfloor x^2 - 3 \rfloor$ and

$g(x) = |x|f(x) + |4x - 7|f(x)$, where $\lfloor y \rfloor$ denotes the greatest integer less than or equal to y for $y \in \mathbb{R}$. Then

[JEE(Advanced)-2016]

(1) f is discontinuous exactly at three points in $\left[-\frac{1}{2}, 2\right]$

(2) f is discontinuous exactly at four points in $\left[-\frac{1}{2}, 2\right]$

(3) g is **NOT** differentiable exactly at four points in $\left(-\frac{1}{2}, 2\right)$

(4) g is **NOT** differentiable exactly at five points in $\left(-\frac{1}{2}, 2\right)$

Sol. Answer (2, 4)

$$\text{Given, } f(x) = \lfloor x^2 - 3 \rfloor$$

$$g(x) = |x|f(x) + |4x - 7|f(x)$$

$$= |x| \lfloor x^2 - 3 \rfloor + |4x - 7| \lfloor x^2 - 3 \rfloor$$

$$= |x| \lfloor x^2 \rfloor - 3|x| + |4x - 7| \lfloor x^2 \rfloor - 3|4x - 7|$$

Discontinuous at $x = 1, \sqrt{2}, \sqrt{3}, 2$ not differentiable $0, 1, \frac{7}{4}, \sqrt{2}, \sqrt{3}$

15. Let $f: \mathbb{R} \rightarrow (0, 1)$ be a continuous function. Then, which of the following function(s) has(have) the value zero at some point in the interval $(0, 1)$? [JEE(Advanced)-2017]

(1) $x - \int_0^{\frac{\pi}{2}-x} f(t) \cos t \, dt$

(2) $x^9 - f(x)$

(3) $e^x - \int_0^x f(t) \sin t \, dt$

(4) $f(x) + \int_0^{\frac{\pi}{2}} f(t) \sin t \, dt$

Sol. Answer (1, 2)

$$g(x) = x - \int_0^{\frac{\pi}{2}-x} f(t) \cos t \, dt$$

$$g(0) = 0 - \int_0^{\frac{\pi}{2}} f(t) \cos t \, dt < 0 \text{ as } 0 < f(t) \cos t < 1$$

$$g(1) = 1 - \int_0^{\frac{\pi}{2}-1} f(t) \cos t \, dt > 0$$

$$g(x) = x^9 - f(x) \rightarrow g(0) = 0 - f(0) < 0$$

$$g(1) = 1 - f(1) > 0$$

Also for $(0, 1)$

$$e^x - \int_0^x f(t) \sin t \, dt > 0$$

$$\text{Also, } f(x) + \int_0^{\pi/2} f(t) \sin t \, dt > 0 \quad \forall x \in (0, 1)$$

SECTION - C

Linked Comprehension Type Questions

Comprehension I

A function $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfies the following conditions

(i) $f(x) \neq 0 \quad \forall x \in \mathbb{R}$

(ii) $f(x+y) = f(x)f(y) \quad \forall x, y \in \mathbb{R}$

(iii) $f(x)$ is differentiable

(iv) $f'(0) = 2$

1. The derivative of $f(x)$ satisfies the equation

(1) $f'(x+y) = f'(x) + f'(y)$

(2) $f'(x+y) = f'(x) \times f'(y)$

(3) $f'(x+y) = f'(x)f'(y)$

(4) $f'(x+y) = f'(x) + f'(y)$

2. The value of $f(0)$ is

(1) 1

(2) -1

(3) 2

(4) $\frac{1}{2}$

3. $\lim_{x \rightarrow 0} \frac{f(x) - f(-x)}{x}$

(1) 1

(2) 2

(3) 3

(4) 4

Solutions of Comprehension I

$$f: R \rightarrow R, f(x) \neq 0, x \in R$$

$$f(x+y) = f(x)f(y), x, y \in R$$

$f(x)$ is differentiable

$$f'(0) = 2$$

1. Answer (3)

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x)f(h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} f(x) \left[\frac{f(h) - 1}{h} \right] \\ &= f(x) \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} \\ &= f(x) \cdot f'(0) = f'(x) = 2f(x), f'(y) = 2f(y) \end{aligned}$$

$$\Rightarrow f'(x+y) = 2f(x+y)$$

$$\Rightarrow f'(x+y) = 2f(x)f(y) = 2f(y) \frac{f'(x)}{2}$$

$$\therefore f'(x+y) = f'(x)f(y)$$

2. Answer (1)

$$f(x+y) = f(x)f(y)$$

Putting $x = 0, y = 0$ we get

$$f(0) = [f(0)]^2$$

$$\Rightarrow f(0)[f(0) - 1] = 0$$

$$\Rightarrow f(0) = 1 \text{ as } f(0) \neq 0$$

3. Answer (4)

$$f'(x) = 2f(x) \Rightarrow \frac{f'(x)}{f(x)} = 2$$

$$\Rightarrow \ln f(x) = 2x + c, \text{ on integration, where } c \text{ is a constant of integration}$$

Putting $x = 0$, we get

$$\ln f(0) = 2 \times 0 + c \Rightarrow c = 0$$

$$\therefore \ln f(x) = 2x \Rightarrow f(x) = e^{2x}$$

$$\text{Now } \lim_{x \rightarrow 0} \frac{f(x) - f(-x)}{x} = \lim_{x \rightarrow 0} \frac{e^{2x} - e^{-2x}}{x}$$

$$\text{Applying L. Hospital rule} = \lim_{x \rightarrow 0} 2e^{2x} + 2e^{-2x} = 4$$

Comprehension II

If $\lim_{x \rightarrow a} \phi(x) = 1$ and $\lim_{x \rightarrow a} \phi(x) = \infty$. Then $\lim_{x \rightarrow a} \{\phi(x)\}^{g(x)} = e^{\lim_{x \rightarrow a} (\phi(x)-1)g(x)}$.

1. The value of $f(0)$ so that the function $f(x) = \left(\frac{2^{2x} + 2^{3x} + 2^{4x} + 2^{5x}}{4} \right)^{1/x}$, $x \neq 0$, continuous at $x = 0$, is given by

- (1) $2^{\frac{7}{4}}$ (2) 2^7 (3) $2^{\frac{7}{2}}$ (4) 2^{14}

Sol. Answer (3)

For continuous function L.H.L. = R.H.L. = $f(0)$

$$\lim_{x \rightarrow 0} \left(\frac{4^x + 8^x + 16^x + 32^x}{4} \right)^{\frac{1}{x}} = 1^\infty$$

$$e^{\lim_{x \rightarrow 0} \frac{1}{4} \left[\frac{4^x - 1}{x} + \frac{8^x - 1}{x} + \frac{16^x - 1}{x} + \frac{32^x - 1}{x} \right]}$$

$$e^{\lim_{x \rightarrow 0} \frac{1}{4} [\log(4 \times 8 \times 16 \times 32)]}$$

$$e^{\lim_{x \rightarrow 0} \frac{1}{4} \log_e(2^{14})} = e^{\frac{7}{2} \log_e 2} = 2^{\frac{7}{2}}$$

2. The value of $f(0)$, so that the function $f(x) = \left(\frac{\sin x}{x} \right)^{\frac{x \sin x}{x^2 - \sin^2 x}}$, $x \neq 0$ is continuous at $x = 0$ is

- (1) e^2 (2) $e^{\frac{1}{2}}$ (3) e (4) 1

Sol. Answer (2)

$$e^{\lim_{x \rightarrow 0} \frac{x \sin x}{x^2 - \sin^2 x} \left(\frac{\sin x}{x} - 1 \right)}$$

$$= e^{\lim_{x \rightarrow 0} \frac{x \sin x (\sin x - x)}{(x - \sin x)(x + \sin x)x}}$$

$$= e^{\lim_{x \rightarrow 0} \frac{-\sin x}{x + \sin x}} = e^{-\frac{1}{2}}$$

Comprehension III

Let $f: R \rightarrow R$ be a differential function satisfying $f\left(\frac{x+y}{3}\right) = \frac{2+f(x)+f(y)}{3}$ for all real x and y and $f'(2) = 2$

1. The function $f(x)$ is equal to

- (1) $\frac{x}{3} + 5$ (2) $2x + 2$ (3) $\frac{x^2 - 2x}{x}$ (4) $\frac{2+x}{3}$

Sol. Answer (2)

$$\begin{aligned}
 \therefore f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{f\left(\frac{3x+3h}{3}\right) - f\left(\frac{3x+0}{3}\right)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\frac{2+f(3x)+f(3h)}{3} - \frac{2+f(3x)+f(0)}{3}}{h} \\
 &= \lim_{h \rightarrow 0} \frac{f(3h) - f(0)}{3h - 0} = f'(0) \\
 \Rightarrow f'(2) &= f'(0) = 2 \quad (\because f'(2) = 2) \\
 \Rightarrow f'(x) &= 2 \Rightarrow f(x) = 2x + 2
 \end{aligned}$$

2. The number of solutions of the equation

$$x^2 + (f(|x|))^2 = 9$$

(1) 3

(2) 5

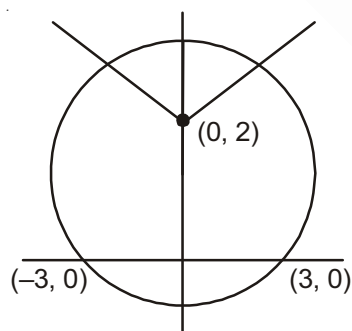
(3) 2

(4) 4

Sol. Answer (3)

$$\text{Let } y = f(|x|) = 2(|x| + 2)$$

$$\text{then } x^2 + y^2 = 3^2$$

 $\therefore$ Number of solutions = 2**SECTION - D****Assertion-Reason Type Questions**

1. STATEMENT-1 : $f(x) = \frac{1 - \cos(1 - \cos x)}{x^4}$ is continuous if $f(0) = \frac{1}{8}$.

and

STATEMENT-2 : $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^-} f(x) = \frac{1}{8}$.

Sol. Answer (1)

$$\text{Statement-1 : } f(0) = \lim_{x \rightarrow 0} \frac{1 - \cos(1 - \cos x)}{x^4}$$

$$= \lim_{x \rightarrow 0} \frac{1 - \cos\left(2\sin^2 \frac{x}{2}\right)}{x^4} = \lim_{x \rightarrow 0} \frac{2\sin^2\left(\sin^2 \frac{x}{2}\right)}{x^4} = \lim_{x \rightarrow 0} \frac{2}{16} \left[\frac{\sin\left(\sin^2 \frac{x}{2}\right)}{\left(\sin^2 \frac{x}{2}\right)} \right]^2 \cdot \left(\frac{\sin \frac{x}{2}}{\frac{x}{2}} \right)^4 = \frac{1}{8}$$

$\Rightarrow$ Statement-1 is true.

Statement-2 : LHL = RHL.

Statement 1 is true Statement 2 is true, Statement-2 is correct explanation of Statement-1.

2. STATEMENT-1 : If $f(x) = \log_{x^2}(\log x)$, then $f'(e) = \frac{1}{e}$.

and

STATEMENT-2 : If $a > 0$, $b > 0$ and $a \neq b$ then $\log_a b = \frac{\log b}{\log a}$.

Sol. Answer (4)

$$\text{Statement-1 : } f(x) = \log_{x^2}(\log x) = \frac{\log(\log x)}{2\log x}$$

$$f'(x) = \frac{\frac{1}{x} \cdot \frac{\log(\log x)}{x}}{2(\log x)^2}$$

$$f'(e) = \frac{1}{2e}$$

Statement-1 is false.

Statement-2 : is true.

3. STATEMENT-1 : Let f be a twice differentiable function such that $f'(x) = g(x)$ and $f''(x) = -f(x)$. If $h'(x) = [f(x)]^2 + [g(x)]^2$, $h(1) = 8$ and $h(0) = 2 \Rightarrow h(2) = 14$.

and

STATEMENT-2 : $h''(x) = 0$.

Sol. Answer (2)

$$\text{Statement-1 : } f(x) = g(x) \Rightarrow f'(x) = g'(x)$$

$$f'(x) = -f(x)$$

$$\text{Also } h'(x) = [f(x)]^2 + [g(x)]^2$$

$$\Rightarrow h''(x) = 2f(x)f'(x) + 2g(x)g'(x) = 2f(x)g(x) + 2g(x)f''(x) = 2f(x)g(x) - 2g(x)f(x) = 0$$

$$\Rightarrow h'(x) = a, a \text{ is a constant of integration} \Rightarrow h(x) = ax + b \Rightarrow h(1) = a + b = 8$$

$$\Rightarrow h(0) = b = 2 \Rightarrow a = 6. \Rightarrow h(x) = 6x + 2 \Rightarrow h(2) = 14$$

i.e., Statement-1 is true.

Statement-2 : $h''(x)$ is true. Statement-2 is not a correct explanation of statement-1

4. STATEMENT-1 : For the function $y = f(x)$, $\frac{\left\{1 + \left(\frac{dy}{dx}\right)^2\right\}^{\frac{3}{2}}}{\frac{d^2y}{dx^2}} = -\frac{\left\{1 + \left(\frac{dx}{dy}\right)^2\right\}^{\frac{3}{2}}}{\frac{d^2x}{dy^2}}$, where $f(x)$ is twice differentiable function.

and

STATEMENT-2 : $\frac{dy}{dx} = \frac{1}{\frac{dx}{dy}}$ and $\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right)$.

Sol. Answer (1)

Statement-1 : $y = f(x)$

Differentiating it w.r.t. x

$$\frac{dy}{dx} = f'(x)$$

$$\frac{d^2y}{dx^2} = f''(x)$$

Differentiating again w.r.t. y , we get

$$1 = f'(x) \frac{dx}{dy}$$

$$\frac{dx}{dy} = \frac{1}{f'(x)}$$

$$\frac{d^2x}{dy^2} = -\frac{1}{f'(x)^2} \cdot f''(x) \cdot \frac{dx}{dy} = -\frac{f''(x)}{\{f'(x)\}^3}$$

Now $\frac{\left\{1 + \left(\frac{dy}{dx}\right)^2\right\}^{\frac{3}{2}}}{\frac{d^2y}{dx^2}} = \frac{\left\{1 + (f'(x))^2\right\}^{\frac{3}{2}}}{f''(x)} \dots (i)$

and $\frac{\left\{1 + \left(\frac{dx}{dy}\right)^2\right\}^{\frac{3}{2}}}{\frac{d^2x}{dy^2}} = \frac{\left\{1 + \frac{1}{(f'(x))^2}\right\}^{\frac{3}{2}}}{-\frac{f''(x)}{(f'(x))^3}} = -\frac{\{1 + (f'(x))^2\}^{\frac{3}{2}} \cdot (f'(x))^3}{(f'(x))^3 f''(x)}$

$$= -\frac{\{1 + (f'(x))^2\}^{\frac{3}{2}}}{f''(x)} \dots (ii)$$

From (i) and (ii) it follows that

$$\frac{\left\{1 + \left(\frac{dy}{dx}\right)^2\right\}^{\frac{3}{2}}}{\frac{d^2y}{dx^2}} = -\frac{\left\{1 + \left(\frac{dx}{dy}\right)^2\right\}^{\frac{3}{2}}}{\frac{d^2x}{dy^2}}$$

i.e., Statement-1 is true.

Statement-2 : We know $\frac{dy}{dx} = \frac{1}{\frac{dx}{dy}}$ and $\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right)$

Statement-2 is true and it is a correct explanation of Statement-1.

5. STATEMENT-1 : $\lim_{x \rightarrow 0} \frac{\sqrt{1 - \cos 2x}}{x}$ does not exist.

and

STATEMENT-2 : Right hand limit $\neq$ Left hand limit.

Sol. Answer (1)

Statement-1 : $\lim_{x \rightarrow 0} \frac{\sqrt{1 - \cos 2x}}{x}$

$$\lim_{x \rightarrow 0} \frac{\sqrt{2} |\sin x|}{x}$$

We observe that

$$\text{LHL} = -\sqrt{2}$$

$$\text{RHL} = \sqrt{2}$$

$$\therefore \text{LHL} \neq \text{RHL}$$

$\Rightarrow$ Limit does not exist.

Statement-2 : is true.

Statement 1 is true Statement 2 is true, Statement-2 is a correct explanation of Statement-1.

6. STATEMENT-1 : If $f(x)$ is differentiable increasing function then $\lim_{x \rightarrow 0} \frac{f(x^2) - f(x)}{f(x) - f(0)} = -1$.

and

STATEMENT-2 : $f(x)$ is an increasing function $\Rightarrow f'(x)$ is positive.

Sol. Answer (2)

Statement-1 : $\lim_{x \rightarrow 0} \frac{f(x^2) - f(x)}{f(x) - f(0)}$

Applying L. Hospital rule

$$\lim_{x \rightarrow 0} \frac{2xf'(x^2) - f'(x)}{f'(x)} = -1$$

i.e., Statement-1 is true.

Statement-2 : If $f(x)$ is increasing function then $f'(x) > 0$

i.e., Statement-2 is true but it is not correct explanation of statement-1.

7. STATEMENT-1 : $f(x) = \cos x$ then $-1 \leq f(x) \leq 1 \Rightarrow -1 \leq f'(x) \leq 1$.

and

STATEMENT-2 : If $f(x) \in [a, b] \Rightarrow f'(x) \in [a, b]$.

Sol. Answer (3)

$$\text{Statement-1 : } f(x) = \cos x \Rightarrow -1 \leq \cos x \leq 1$$

$$f'(x) = -\sin x \Rightarrow -1 \leq f'(x) \leq 1$$

Statement-1 is true.

Statement-2 : If $f(x) \in [a, b]$ then it is not necessary

$$f'(x) \in [a, b]$$

Statement-2 is false.

SECTION - E**Matrix-Match Type Questions**1. For f to be continuous at $x = 0$ $f(0)$ is given by**Column-I**

$$(A) f(x) = \frac{\ln(1+4x)}{x}$$

$$(B) f(x) = \frac{\ln(4+x) - \ln 4}{x}$$

$$(C) f(x) = \frac{1}{\sin x} - \frac{1}{\tan x}$$

$$(D) f(x) = \frac{1 - \cos^3 x}{x \sin 2x}$$

Column-II

$$(p) \frac{1}{4}$$

$$(q) 0$$

$$(r) 4$$

$$(s) \frac{3}{4}$$

Sol. Answer A(r), B(p), C(q), D(s)

$$(A) f(0) = \lim_{x \rightarrow 0} \frac{\ln(1+4x)}{x} = 4$$

$$(B) f(0) = \lim_{x \rightarrow 0} \frac{\ln(4+x) - \ln 4}{x} = \frac{1}{4}$$

$$(C) f(0) = \lim_{x \rightarrow 0} \left(\frac{1}{\sin x} - \frac{1}{\tan x} \right) = \frac{1 - \cos x}{\sin x} \times \frac{1 + \cos x}{1 + \cos x} = \lim_{x \rightarrow 0} \frac{\sin x}{1 + \cos x} = 0$$

$$(D) f(0) = \lim_{x \rightarrow 0} \frac{1 - \cos^3 x}{x \sin 2x} = \lim_{x \rightarrow 0} \frac{1 - \cos^3 x}{2x^2 \frac{\sin 2x}{2x}} = \lim_{x \rightarrow 0} \frac{1 - \cos^3 x}{2x^2}$$

Applying L. Hospital rule

$$\lim_{x \rightarrow 0} \frac{3 \cos^2 x \cdot \sin x}{4x} = \frac{3}{4}$$

2. Match the following

Column I

Column II

(A) If $f(x) = \begin{cases} x^k \cos \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$ is continuous at $x = 0$, then (p) 2

the least integral value that k can attain is

(B) If $f(x) = (x+1)^{\cot x}$ is continuous at $x = 0$ and $f(0) = k$, then $[k]$ is ([.] is G.I.F.) (q) A rational number

(C) $\frac{d^4}{dx^4}(x^3 \log x) = \frac{k}{x}$ then k is (r) 1

(D) If $f(x+y) = f(x)f(y) \forall x, y$ and $f(k) = 2$, for some $k \in R$, $f'(0) = 3$, then $f'(k)$ is (s) 6

(t) An odd number

Sol. Answer A(q, r, t), B(p, q), C(q, s), D(q, s)

(A) $\lim_{x \rightarrow 0} x^k \cos \frac{1}{x} = 0$ if $k > 1$.

Hence the least integral value that k can attain for the function $f(x)$ to be continuous at $x = 0$ is 1.

1 is rational number. Also 1 is odd

(B) Since $f(x)$ is continuous at $x = 0$

$$k = f(0) = \lim_{x \rightarrow 0} (1+x)^{\cot x} = e^{\lim_{x \rightarrow 0} x \cot x} = e^{\lim_{x \rightarrow 0} \left(\frac{x}{\tan x} \right)} = e^1 = e$$

Now $2 < e < 3 \Rightarrow [k] = [e] = 2$ which a rational number.

(C) $\frac{k}{x} = \frac{d^4}{dx^4}(x^3 \log x) = \frac{d^3}{dx^3}(x^2 + 3x^2 \cdot \log x)$

$$= \frac{d^3}{dx^3}(3x^2 \log x) \left[\frac{d^3}{dx^3}(x^2) = 0 \right]$$

$$= 3 \frac{d^2}{dx^2} \left[\frac{d}{dx}(x^2 \log x) \right]$$

$$= 3 \frac{d^2}{dx^2} [x + 2x \ln x]$$

$$= 6 \frac{d^2}{dx^2} (x \ln x)$$

$$= 6 \frac{d}{dx} [1 + \ln x] = \frac{6}{x}$$

$$\Rightarrow k = 6$$

$$(D) f(x+y) = f(x) + f(y) \Rightarrow f(0) = 1$$

$$\text{Now } f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = f(x) \lim_{h \rightarrow 0} \frac{f(h) - 1}{h} = f(x) \cdot f'(0) = 3f(x)$$

$$\therefore f'(5) = 3f(5) = 3.2 = 6$$

which is a rational number.

3. Match the following :

Column-I

$$(A) 4x^2 + 9y^2 = 36$$

$$(B) x^2 + y^2 = 4$$

$$(C) y = \tan(x+y)$$

$$(D) x^3 + y^3 - 6xy = 0$$

Column-II

$$(p) y'' = -\frac{16xy}{(y^2 - 2x)^3}$$

$$(q) y''' = -\frac{2(3y^4 + 8y^2 + 5)}{y^8}$$

$$(r) y''' = -\frac{12x}{y^5}$$

$$(s) y'' = -\frac{16}{9y^3}$$

Sol. Answer A(s), B(r), C(q), D(p)

$$(A) 4x^2 + 9y^2 = 36$$

$$\Rightarrow 4x + 9yy' = 0$$

$$\Rightarrow y' = -\frac{4x}{9y}$$

$$\Rightarrow 4 + 9yy'' + 9(y')^2 = 0$$

$$\Rightarrow 9yy'' = -4 - \frac{16x^2}{9y^2}$$

$$\Rightarrow 9yy'' = -\frac{4(4x^2 + 9y^2)}{9y^2}$$

$$\Rightarrow y'' = -\frac{16}{9y^3}$$

$$(B) x^2 + y^2 = 4$$

$$\Rightarrow x + yy' = 0$$

$$\Rightarrow 1 + yy'' + (y')^2 = 0$$

$$\Rightarrow yy'' = -1 - \frac{x^2}{y^2} = -\frac{x^2 + y^2}{y^2}$$

$$\Rightarrow y'' = -\frac{4}{y^3}$$

$$\Rightarrow y''' = \frac{12}{y^4} y'$$

$$\Rightarrow y''' = -\frac{12x}{y^5}$$

(C) $y = \tan(x + y)$

$$\Rightarrow y' = \sec^2(x + y)(1 + y')$$

$$\Rightarrow y' = -\frac{1 + y^2}{y^2}$$

$$\Rightarrow y'' = +\frac{2}{y^3} \cdot y' = -\frac{2(1 + y^2)}{y^5}$$

$$\Rightarrow y''' = -2 \frac{2y^6 - (1 + y^2)5y^4}{y^{10}} \cdot y'$$

$$\Rightarrow y''' = -\frac{2(3y^4 + 8y^2 + 5)}{y^8}$$

(D) $x^3 + y^3 - 6xy = 0$

$$\Rightarrow x^2 + y^2 y' - 2(xy' + y) = 0$$

$$\Rightarrow x^2 + y'(y^2 - 2x) - 2y = 0$$

$$\Rightarrow 2x + 2y(y')^2 + (y^2 - 2x)y'' - 4y' = 0$$

$$\Rightarrow y'' = \frac{4y' - 2y(y')^2 - 2x}{(y^2 - 2x)}$$

$$\Rightarrow y'' = -\frac{16xy}{(y^2 - 2x)^3}$$

4. Match the following :

Column-I

(A) $f(x) = \frac{1}{\sqrt{x-2}}$

Column-II

(p) $\lim_{x \rightarrow 0} f(x) = 0$

$$(B) f(x) = \frac{x - \sin x}{x + \sin x}$$

(q) f is continuous on R .

$$(C) f(x) = x \sin \frac{\pi}{x}, f(0) = 0$$

$$(r) \lim_{x \rightarrow \infty} f(x) = 0$$

$$(D) f(x) = \tan^{-1} \frac{1}{x}$$

$$(s) \lim_{x \rightarrow 0} f(x) \text{ does not exist}$$

Sol. Answer A(r, s), B(p), C(p, q), D(r, s)

$$(A) f(x) = \frac{1}{\sqrt{x-2}}$$

$$\Rightarrow \lim_{x \rightarrow \infty} \frac{1}{\sqrt{x-2}} = 0$$

$$(B) \lim_{x \rightarrow 0} \frac{x - \sin x}{x + \sin x}$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{1 - \frac{\sin x}{x}}{1 + \frac{\sin x}{x}} = 0$$

$$(C) f(x) = x \sin \frac{\pi}{x}$$

At $x = 0$

$$\text{LHL} = \lim_{x \rightarrow 0^-} x \sin \frac{\pi}{x} = 0$$

$$\text{RHL} = \lim_{x \rightarrow 0^+} x \sin \frac{\pi}{x} = 0$$

$$f(0) = 0$$

i.e., f is continuous on R .

$$(D) f(x) = \tan^{-1} \frac{1}{x}$$

At $x = 0$

$$\text{RHL} = \lim_{x \rightarrow 0^+} \tan^{-1} \frac{1}{x} = \frac{\pi}{2}$$

$$\text{LHL} = \lim_{x \rightarrow 0^-} \tan^{-1} \frac{1}{x} = -\frac{\pi}{2}$$

$$\text{LHL} \neq \text{RHL}$$

i.e., $\lim_{x \rightarrow 0} f(x)$ does not exist.



5. Let $f_1 : \mathbb{R} \rightarrow \mathbb{R}$, $f_2 : \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \rightarrow \mathbb{R}$, $f_3 : \left(-1, e^{\frac{\pi}{2}} - 2\right) \rightarrow \mathbb{R}$ and $f_4 : \mathbb{R} \rightarrow \mathbb{R}$ be functions defined by

$$(i) f_1(x) = \sin\left(\sqrt{1 - e^{-x^2}}\right)$$

$$(ii) f_2(x) = \begin{cases} \frac{|\sin x|}{\tan^{-1} x} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases}, \text{ where the inverse trigonometric function } \tan^{-1} x \text{ assumes values in } \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

(iii) $f_3(x) = [\sin(\log_e(x+2))]$, where, for $t \in \mathbb{R}$, $[t]$ denotes the greatest integer less than or equal to t ,

$$(iv) f_4(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

[JEE(Advanced)-2018]

Column-I

(A) The function f_1 is

(B) The function f_2 is

(C) The function f_3 is

(D) The function f_4 is

Column-II

(p) NOT continuous at $x = 0$

(q) Continuous at $x = 0$ and NOT differentiable at $x = 0$

(r) Differentiable at $x = 0$ and its derivative is NOT continuous at $x = 0$

(s) Differentiable at $x = 0$ and its derivative is continuous at $x = 0$

The correct option is

(1) A(q); B(r); C(p); D(s)

(2) A(s); B(p); C(q); D(r)

(3) A(s); B(q); C(p); D(r)

(4) A(q); B(p); C(s); D(r)

Sol. Answer (4)

(i) $f_1 : \mathbb{R} \rightarrow \mathbb{R}$

$$f_1(x) = \sin\left(\sqrt{1 - e^{-x^2}}\right) = \sin\left(\sqrt{1 - \frac{1}{e^{x^2}}}\right)$$

$f_1(x)$ is continuous everywhere, but $f_1'(0)$ does not exist, therefore $f_1(x)$ is not differentiable function at $x = 0$.

(ii) $f_2 : \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \rightarrow \mathbb{R}$

$$f_2(x) = \begin{cases} -\frac{\sin x}{\tan^{-1} x} & ; x < 0 \\ 1 & ; x = 0 \\ \frac{\sin x}{\tan^{-1} x} & ; x > 0 \end{cases}$$

LHL at $x = 0$ is -1

RHL at $x = 0$ is 1

$\therefore f_2(x)$ is discontinuous at $x = 0$,

$$(iii) f_3: \left(-1, e^{\frac{\pi}{2}} - 2\right) \rightarrow R$$

$$f_3(x) = [\sin(\log_e(x+2))]$$

$$\therefore -1 < x < e^{\frac{\pi}{2}} - 2$$

$$1 < x + 2 < e^{\frac{\pi}{2}}$$

$$\Rightarrow 0 < \log_e(x+2) < \frac{\pi}{2}$$

$$\Rightarrow 0 < \sin(\log_e(x+2)) < 1$$

$$\Rightarrow [\sin(\log_e(x+2))] = 0$$

$$\Rightarrow f_3(x) = 0$$

$\Rightarrow f_3(x)$ is continuous and differentiable at $x = 0$

$$(iv) f_4: R \rightarrow R$$

$$f_4(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right) & ; x \neq 0 \\ 0 & ; x = 0 \end{cases}$$

$$\lim_{x \rightarrow 0} f_4(x) = \lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right) = 0$$

$$= \text{Value of } f_4(x) \text{ at } x = 0$$

$\Rightarrow f_4(x)$ is continuous at $x = 0$

$$\text{Now, } f'(0^+) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{\sin\left(\frac{1}{h}\right)}{\frac{1}{h}} = 0$$

$$f'(0^-) = \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{-h} = \lim_{h \rightarrow 0} \frac{\sin\left(\frac{1}{h}\right)}{\frac{1}{h}} = 0$$

$\Rightarrow f(x)$ is differentiable at $x = 0$

$$\text{Now, } f'_4(x) = 2x \sin\left(\frac{1}{x}\right) - x^2 \cdot \frac{1}{x^2} \cos\left(\frac{1}{x}\right)$$

$$= 2x \sin\left(\frac{1}{x}\right) - \cos\left(\frac{1}{x}\right) \text{ is oscillating}$$

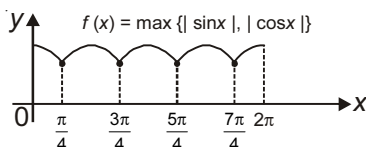
Function which discontinuous at $x = 0$

SECTION - F

Integer Answer Type Questions

1. The number of points where $f(x) = \max\{|\sin x|, |\cos x|\}$, $x \in (-2\pi, 2\pi)$ is not differentiable is _____.

Sol. Answer (8)

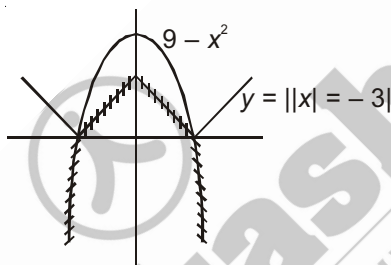


The function is even. There are four points in $(0, 2\pi)$ where it is not differentiable. Hence in $[-2\pi, 2\pi]$ there are 8 such points

2. The number of points where $g(x) = \min\{|x-3|, 9-x^2\}$ ($x \in \mathbb{R}$) is not differentiable is _____.

Sol. Answer (3)

It is clear from the graph that there are three points where there is sharp corner. Hence there are three points where function is not differentiable.



3. If the right hand derivative of $h(x) = \{x\}$ ($\{.\}$ is fractional part of x) exists at $x = 1$ and it is equal to _____.

Sol. Answer (1)

$$\lim_{h \rightarrow 0} \frac{\{1+h\} - \{1\}}{h} = \lim_{h \rightarrow 0} \frac{h-0}{h} = \lim_{h \rightarrow 0} (1) = 1$$

4. If $y = f(x)$ and $y^4 - 4y + x = 0$. If $f(-8) = 2$ then the value of $|28 f'(-8)|$ is _____.

Sol. Answer (1)

$$y^4 - 4y + x = 0$$

$$\text{at } y = 2, x = 8$$

Differentiating the equation

$$4(y^3 - 1) \frac{dy}{dx} + 1 = 0$$

$$\frac{dy}{dx} = \frac{-1}{4(y^3 - 1)}$$

$$\Rightarrow f'(x) = -\frac{1}{4(y^3 - 1)}$$

$$f'(-8) = -\frac{1}{4(2^3 - 1)} = -\frac{1}{28}$$

$$\Rightarrow |28 f'(-8)| = 1$$

5. If $f(x) = f(1-x)$ and $f(x)$ is differentiable at every real value of x then the value of $f'\left(\frac{1}{2}\right) + f'\left(\frac{1}{4}\right) + f'\left(\frac{3}{4}\right)$ is ____.

Sol. Answer (0)

$$f(x) = f(1-x)$$

$$\Rightarrow f'(x) = -f'(1-x)$$

$$\text{at } x = \frac{1}{2}$$

$$f'\left(\frac{1}{2}\right) = -f'\left(1 - \frac{1}{2}\right)$$

$$f'\left(\frac{1}{2}\right) + f'\left(\frac{1}{2}\right) = 0$$

$$\Rightarrow f'\left(\frac{1}{2}\right) = 0 \quad \dots(i)$$

$$\text{Also } x = \frac{1}{4}$$

$$f'\left(\frac{1}{4}\right) = -f'\left(\frac{3}{4}\right)$$

$$\Rightarrow f'\left(\frac{1}{4}\right) + f'\left(\frac{3}{4}\right) = 0 \quad \dots(ii)$$

By (i) and (ii)

$$f'\left(\frac{1}{2}\right) + f'\left(\frac{1}{4}\right) + f'\left(\frac{3}{4}\right) = 0$$

6. If $\lim_{x \rightarrow 0} \left[\frac{m \sin x}{x} \right] = 8$ then the value of m is ($[.]$ in the greatest integer function).

Sol. Answer (9)

We know that

$$\frac{\sin x}{x} < 1$$

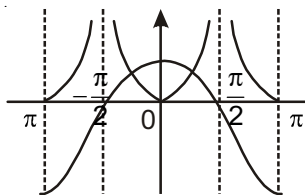
$$\Rightarrow \frac{m \sin x}{x} < m \Rightarrow \lim_{x \rightarrow 0} \left[\frac{m \sin x}{x} \right] = (m-1)$$

$$\text{But given that } \lim_{x \rightarrow 0} \left[\frac{m \sin x}{x} \right] = 8$$

$$\text{so } m-1 = 8 \Rightarrow m = 9$$

7. The number of points, where the function $f(x) = \max(|\tan x|, |\cos x|)$ is non-differentiable in the interval $(-\pi, \pi)$ is

Sol. Answer (4)



The function is not differentiable and continuous at two points between $x = -\frac{\pi}{2}$ and $x = \frac{\pi}{2}$. Also the function is not continuous at $x = \frac{\pi}{2}$ and $x = -\frac{\pi}{2}$. Hence at four points function is not differentiable.

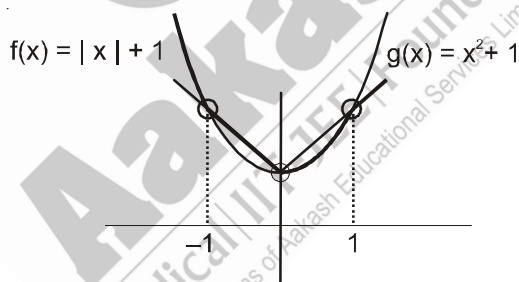
8. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ be respectively given by $f(x) = |x| + 1$ and $g(x) = x^2 + 1$. Define $h: \mathbb{R} \rightarrow \mathbb{R}$ by

$$h(x) = \begin{cases} \max \{f(x), g(x)\} & \text{if } x \leq 0, \\ \min \{f(x), g(x)\} & \text{if } x > 0. \end{cases}$$

The number of points at which $h(x)$ is not differentiable is

[JEE(Advanced)-2014]

Sol. Answer (3)



9. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous odd function, which vanishes exactly at one point and $f(1) = \frac{1}{2}$. Suppose that

$$F(x) = \int_{-1}^x f(t) dt \text{ for all } x \in [-1, 2] \text{ and } G(x) = \int_{-1}^x t |f(t)| dt \text{ for all } x \in [-1, 2]. \text{ If } \lim_{x \rightarrow 1} \frac{F(x)}{G(x)} = \frac{1}{14}, \text{ then the value}$$

of $f\left(\frac{1}{2}\right)$ is

Sol. Answer (7)

$$f(1) = \frac{1}{2}; f(-x) = -f(x); f(-1) = -\frac{1}{2} \text{ and } f(x) \text{ is zero only at one point}$$

$$F(x) = \int_{-1}^x f(t) dt \quad x \in [-1, 2]$$

$$F'(x) = f(x)$$

$$G(x) = \int_{-1}^x f |f(f(t))| dt$$

$$\Rightarrow G'(x) = x |f(f(x))|$$

$$\lim_{x \rightarrow 1} \frac{F(x)}{G(x)} = \lim_{x \rightarrow 1} \frac{F'(x)}{G'(x)} = \lim_{x \rightarrow 1} \frac{f(x)}{x |f(f(x))|}$$

$$= \frac{\frac{1}{2}}{\left| f\left(\frac{1}{2}\right) \right|} = \frac{1}{14}$$

$$\Rightarrow \left| f\left(\frac{1}{2}\right) \right| = 7$$

$$\Rightarrow f\left(\frac{1}{2}\right) = 7 \text{ as } f\left(\frac{1}{2}\right) \text{ cannot be negative}$$

SECTION - G

Multiple True-False Type Questions

1. STATEMENT-1 : $y = [x]$ ([.] denotes greatest integer function) is not a continuous function.

STATEMENT-2 : $y = \{x\}$ ({.} denotes fractional function) is discontinuous at integral points.

STATEMENT-3 : $y = 7^x$ is continuous in its domain.

(1) T F T

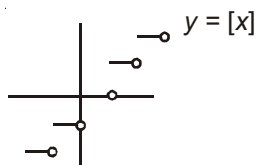
(2) T T T

(3) F F F

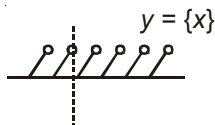
(4) F F T

Sol. Answer (2)

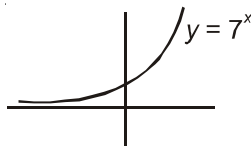
STATEMENT – 1 : True from graph



STATEMENT – 2 : True from graph



STATEMENT – 3 : True from graph



2. STATEMENT-1 : $\lim_{x \rightarrow 0} \frac{\sin x}{x}$ exists.

STATEMENT-2 : $|x|$ is differentiable at $x = 0$.

STATEMENT-3 : If $\lim_{x \rightarrow 0} \frac{\tan kx}{\sin 5x} = 3$, then $k = 15$.

(1) T F T

(2) T T T

(3) F F F

(4) F F T

Sol. Answer (1)

STATEMENT – 1 : True. As, $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$

STATEMENT – 2 : False. The graph of $|x|$ has sharp corner at $x = 0$

STATEMENT – 3 : True $\lim_{x \rightarrow 0} \left(\frac{\frac{\tan kx}{kx}}{\frac{\sin 5x}{5x}} \right) \cdot \frac{kx}{5x} = 3$

$$1. \frac{k}{5} = 3 \Rightarrow k = 15$$

3. STATEMENT-1 : $\lim_{x \rightarrow \infty} \frac{\log[x]}{[x]} = 0$, where $[x]$ represents the integral part of x .

STATEMENT-2 : $\lim_{x \rightarrow 0} \frac{\sqrt{\sec^2 x - 1}}{x}$ does not exist.

STATEMENT-3 : $\lim_{x \rightarrow 2} (x-1)^{\frac{1}{x-2}} = 1$

(1) T T T

(2) T T F

(3) F T F

(4) F F F

Sol. Answer (2)

Statement – 1 : $\lim_{x \rightarrow \infty} \frac{\log[x]}{[x]} = 0$

As $\frac{\ln x}{x}$ is decreasing for $x > 1$

$$\therefore \frac{\ln x}{x} \leq \frac{\ln(x)}{(x)} \leq \frac{\ln(x-1)}{x-1}$$

Using Sandwich theorem

$$\lim_{x \rightarrow \infty} \frac{\ln[x]}{[x]} = 0$$

Statement - 2 : $\lim_{x \rightarrow 0} \frac{[\tan x]}{x}$

$$\text{LHL} = \lim_{x \rightarrow 0-h} \frac{[\tan(-h)]}{-h} = -1$$

$$\text{RHL} = \lim_{x \rightarrow 0+h} \frac{[\tan h]}{h} = 1$$

Statement - 3 : $\lim_{x \rightarrow 2} (1+x-2)^{\frac{1}{x+2}} = e^{\lim_{x \rightarrow 2} \frac{1}{x-2}(x-2)} = e^1$

4. $\lim_{x \rightarrow 0} \frac{A \sin x + B \log(1+x^2) + C(1-\cos x)}{x^2} = 2$, then

STATEMENT-1 : $A = 1$

STATEMENT-2 : $2B + C = 2$

STATEMENT-3 : $A + 2B + C = 5$

(1) T T T

(2) T T F

(3) F T F

(4) F F F

Sol. Answer (4)

$$\lim_{x \rightarrow 0} \frac{A \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} + \dots \right) + B \left(x^2 - \frac{x^4}{2} + \dots \right) + C \left(1 - \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots \right) \right)}{x^2} = 2$$

$$\lim_{x \rightarrow 0} \frac{x(A) + x^2 \left(B + \frac{C}{2} \right) + x^3 \left(-\frac{A}{3!} \right) + \dots}{x^2} = 2$$

$$\Rightarrow A = 0, B + \frac{C}{2} = 2$$

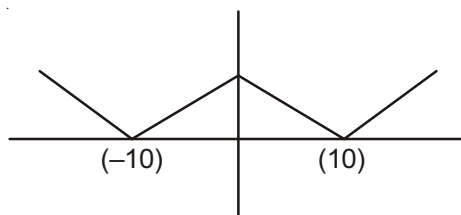
Hence all the statements are false.

SECTION - H

Aakash Challengers Questions

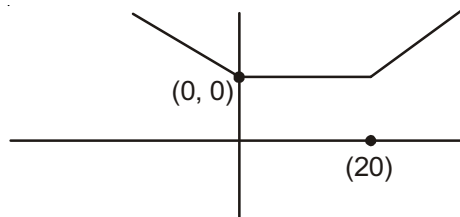
1. Let $f(x) = ||x| - 1|$, $g(x) = |x| + |x - 2|$, $h(x) = \max. \{1, x, x^3\}$. If a, b, c are the no. of points where $f(x)$, $g(x)$ and $h(x)$ are not differentiable, then the value of $a + b + c$ is

Sol. Answer (6)

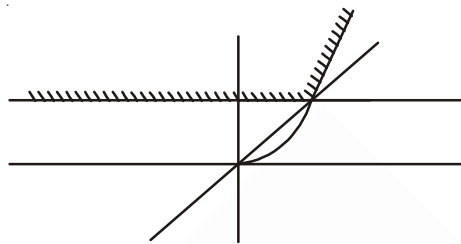


Graph of $f(x) = ||x| - 1|$

$$g(x) = |x| + |x - 2|$$



$$h(x) = \max\{1, x, x^3\}$$



∴ 6 points.

2. Let $P(x)$ be a polynomial of degree 4, with $P(2) = -1$, $P'(2) = 0$, $P''(2) = 2$, $P'''(2) = -12$ and $P^{IV}(2) = 24$. Calculate $P''(1)$

Sol. Let $p(x) = ax^4 + bx^3 + cx^2 + dx + e$

$$\therefore p'(x) = 4ax^3 + 3bx^2 + 2cx + d$$

$$\therefore p''(x) = 12ax^2 + 6bx + 2c$$

$$\therefore p'''(x) = 24ax + 6b$$

$$\therefore p^{IV}(x) = 24a$$

$$\therefore p^{IV}(2) = 24a$$

$$\therefore 24 = 24a \quad \therefore a = 1$$

$$\text{Now } p'''(2) = 48a + 6b$$

$$\Rightarrow -12 = 48 + 6b$$

$$\Rightarrow 6b = -60 \quad \Rightarrow b = -10$$

$$\text{Also } p''(2) = 12a(2)^2 + 12b + 2c$$

$$\Rightarrow 2 = 48 - 120 + 2c$$

$$\Rightarrow -46 + 120 = 2c \quad \Rightarrow c = 37$$

$$\text{Hence } p''(1) = 12a + 6b + 2c = 12 - 60 + 74 = 26$$

3. Let $f(x) = [\sin x + \cos x]$, $[.]$ = greatest integer function, $\forall x \in [0, 2\pi]$

$$g(x) = \frac{1}{\log|x|}$$

$$h(x) = \frac{1}{|x| - 2}$$

If a , b and c are the number of points of discontinuities of $f(x)$, $g(x)$ and $h(x)$ respectively then the value of $a + b + c$ is

Sol. $f(x) = [\sin x + \cos x]$

$$\Rightarrow a = 5, \text{ where } x = \left\{ \frac{\pi}{2}, \frac{3\pi}{4}, \pi, \frac{3\pi}{2}, \frac{7\pi}{4} \right\}$$

$$\therefore g(x) = \frac{1}{\log |x|}$$

$$\Rightarrow b = 3, \text{ where } x = \{-1, 0, 1\}$$

$$\therefore h(x) = \frac{1}{|x| - 2}$$

$$\Rightarrow c = 2, \text{ where } x = \{-2, 2\}$$

$$\therefore a + b + c = 5 + 3 + 2 = 10$$

4. Let $f(x) = [n + p \sin x]$, $\forall x \in [0, \pi]$, $n \in \mathbb{N}$ and p is a prime number

$$g(x) = [\sin x + \cos x], \forall x \in [0, \pi]$$

$$h(x) = [\sqrt{2} \sin x], \forall x \in [-2\pi, 2\pi], \text{ where } [.] = \text{Greatest integer function}$$

If a, b, c are number of points, where $f(x)$ is not differentiable such that $a + b + c = 35$, then the value of p is.....

Sol. Given, $f(x) = [n + p \sin x]$

$\Rightarrow f(x)$ is not differentiable at points,
where $n + p \sin x = \text{an integer} = k(\text{say})$

$$\Rightarrow \sin x = \frac{k - n}{p}$$

$$\text{In } [0, \pi], 0 \leq \sin x \leq 1$$

$$\Rightarrow 0 \leq p \sin x \leq p$$

$$\Rightarrow n \leq n + p \sin x \leq n + p$$

$$\Rightarrow n + p \sin x = n, n + 1, n + 2, \dots, n + p$$

$$\Rightarrow p \sin x = 0, 1, 2, \dots, p$$

$$\Rightarrow \sin x = 0, \frac{1}{p}, \frac{2}{p}, \dots, \frac{p-1}{p}, 1$$

$$\text{Now, } \sin x = 0 \Rightarrow x = 0, \pi$$

$$\sin x = \frac{1}{p} \Rightarrow x = \sin^{-1}\left(\frac{1}{p}\right), \pi - \sin^{-1}\left(\frac{1}{p}\right)$$

.....

.....

$$\sin x = \frac{p-1}{p} \Rightarrow x = \sin^{-1}\left(\frac{p-1}{p}\right), \pi - \sin^{-1}\frac{p-1}{p}$$

$$\sin x = 1 \Rightarrow x = \frac{\pi}{2}$$

Thus, $a = 2p + 1$

Also, $g(x) = [\sin x + \cos x]$

$$= \left[\sqrt{2} \sin \left(x + \frac{\pi}{4} \right) \right]$$

$$\Rightarrow \sqrt{2} \sin \left(x + \frac{\pi}{4} \right) = -1, 0, 1$$

$$\Rightarrow \sin \left(x + \frac{\pi}{4} \right) = -\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}$$

$$\Rightarrow x + \frac{\pi}{4} = \pi, \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}$$

$$\Rightarrow x = \frac{3\pi}{4}, 0, \frac{\pi}{2}, \pi$$

$$\Rightarrow b = 4$$

Also, $h(x) = [\sqrt{2} \sin x]$

$$\Rightarrow \sqrt{2} \sin x = -1, 0, 1$$

$$\Rightarrow \sin x = -\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}$$

$$\Rightarrow x = \frac{5\pi}{4}, \frac{7\pi}{4}, \frac{-3\pi}{4}, \frac{-\pi}{4}, 0, \pi, 2\pi, -\pi, -2\pi, \frac{\pi}{4}, \frac{3\pi}{4}, \frac{-5\pi}{4}, \frac{-7\pi}{4}$$

$$\Rightarrow c = 13$$

Now, $a + b + c = 40$

$$\Rightarrow 2p + 1 + 4 + 13 = 40$$

$$\Rightarrow 2p + 1 = 40 - 17 = 23$$

$$\Rightarrow 2p = 22$$

$$\Rightarrow p = 11$$

